



SPRINT 2 REPORT

COS40006 COMPUTING TECHNOLOGY PROJECT B

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Contribution Summary (This Sprint)

	Contribution	Initiatives	Communication					Respect
Team Member	Finished tasks in time or on time with an acceptable quality	Self-assigned tasks and proactively found solutions to problems	Attended all team meetings	Attended all supervisor meetings	Attended all client meetings	Replied every time within an acceptable timeframe	Kept the team updated about the status	The student respected others and listened to different opinions
Kenneth Hui Hong CHUA	Yes, in case of can't make on time will inform the team	Self-researching, find out problem, provide solution to the team	Yes	Yes	Yes	Yes	As leader of team, always updating the status	Yes, listening to team member opinion
Ivan Jia Wei HOAN	If cannot finish will try adjust the task	Will find problem and solution for the project	Yes	Yes	Yes	Reply as soon as possible	Will notice if have any changes	Try to discuss with others when different opinion
Kelvin Wen Kiong FONG	Built the foundation of the system and made the webpage responsive	Helped with tasks when needed and tried my best to solve problems	Yes	Yes	Yes	Reply ASAP	Regularly informed the team about our progress and any changes in the project	Talk with all team members to share feedback and check on any updates or changes
Nigel Zi Jun LING	Follow the requirement and finish task as soon as possible with acceptable quality	Help teams to solve	Yes	Yes	Yes	Reply as soon as possible	Notice every change in physical meetings and WhatsApp group	Communicate with every member to exchange feedback and follow up changes

Contribution Details (This Sprint)

Kenneth Hui Hong CHUA

During this sprint, my primary responsibility was the development and refinement of the AI Book Recommendation System. The system is powered by Google Gemini LLM, which is integrated into the library platform to act as a reading assistant that recommends books directly from the library catalogue based on user input. To improve the reliability and safety of the AI, I implemented a guardrails method that restricts the AI's responses strictly to book-related queries. This prevents users from misusing the chatbot for unintended purposes and reduces unnecessary API token consumption. The guardrail works by embedding a prompt restriction at the system level, ensuring the AI only responds when the user's input is relevant to book searching, genres, topics, or course units.

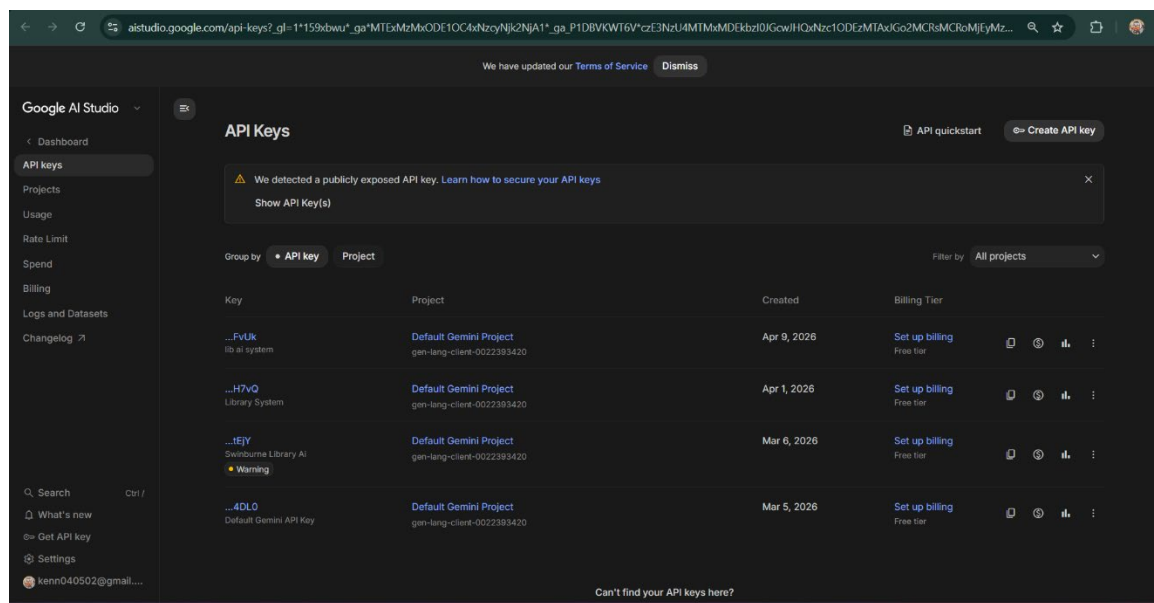


Figure 1: Gemini Api Keys

On the user interface side, I carried out several improvements to the AI Recommendation page to enhance usability across both desktop and mobile devices. These improvements included expanding the chat message panel for better readability, implementing a clear chat history button to allow users to reset their conversation, and optimising the mobile view layout to ensure the chat interface remains functional and accessible on smaller screens.

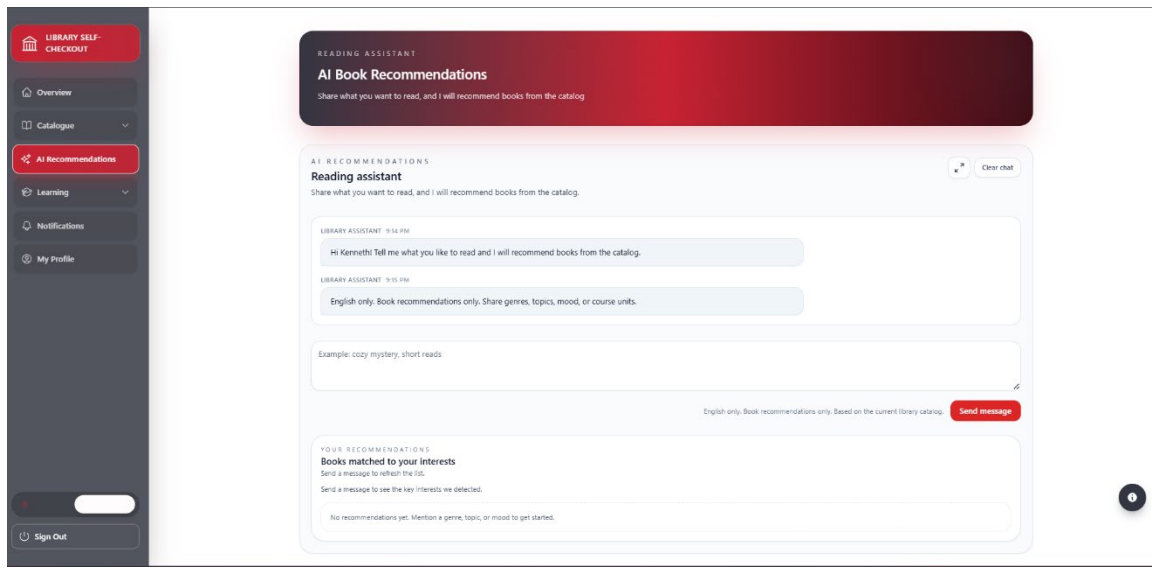


Figure 2: AI Book Recommendation Box (Desktop View)

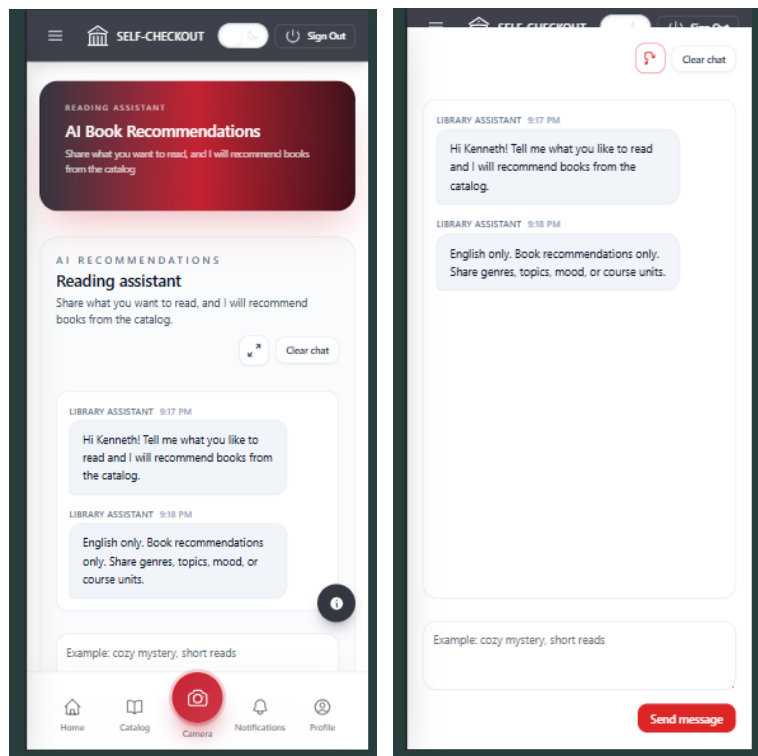


Figure 3: Book Recommendation Box (Mobile View)

For the LinkedIn Learning integration, I explored and attempted to implement a feature that would suggest LinkedIn Learning courses alongside AI book recommendations. However, the integration proved infeasible as LinkedIn Learning requires an institutional license and admin-level API access that was unavailable to the team. As a workaround, I

created a dedicated LinkedIn Learning page within the system that allows users to search for relevant topics and redirects them to the LinkedIn Learning platform to access courses directly.

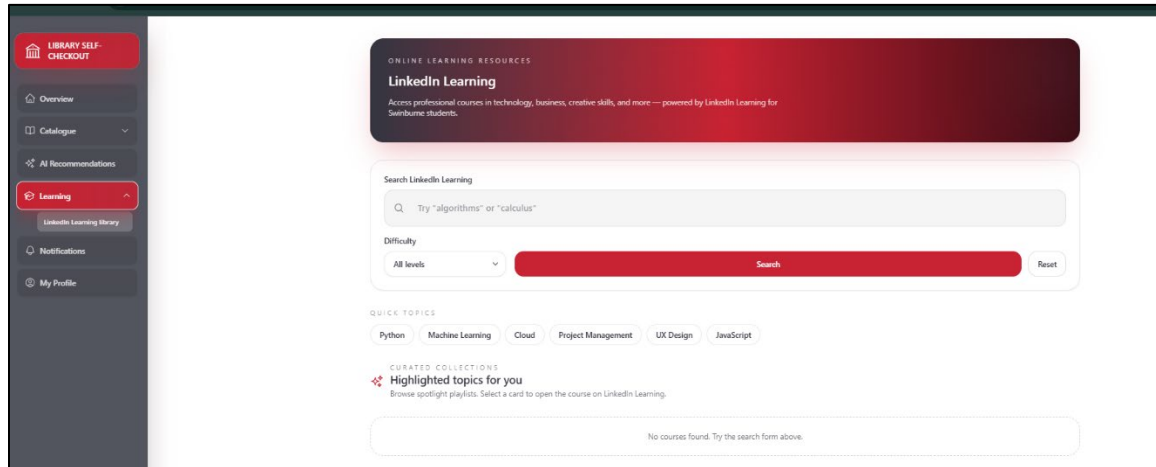


Figure 4: LinkedIn Learning (Desktop View)

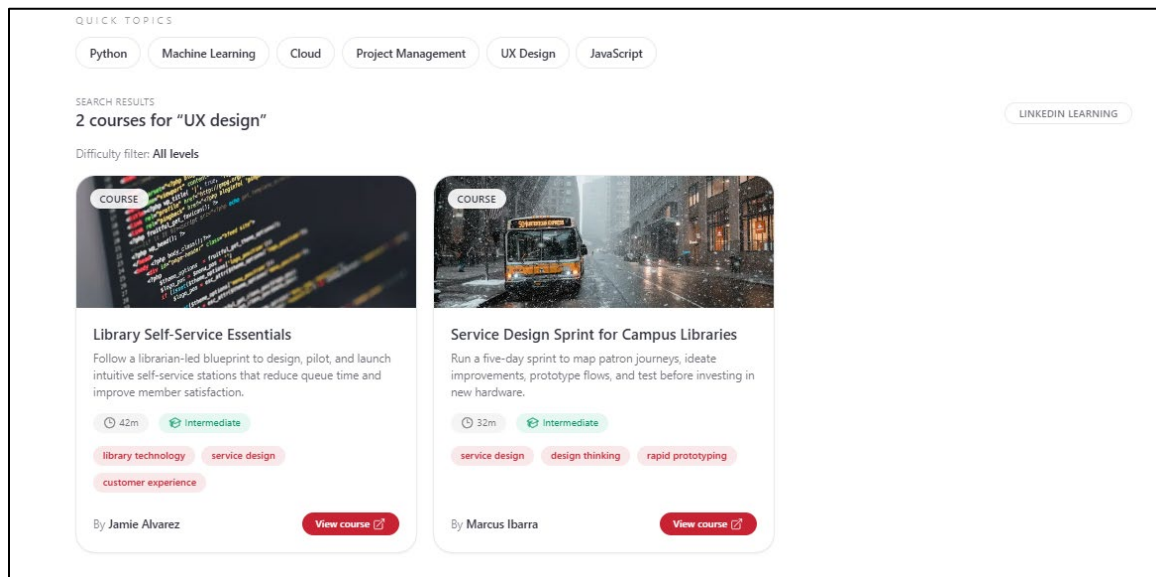


Figure 5: Grib display base on Search Engine

Additionally, I implemented the User Interest feature based on the student's bachelor program. Upon their first login, new users are prompted to select their bachelor program from four available categories: Computer Science, Engineering, Art & Design, and Business. This selection is saved to the database in the UserInterests table and is used by the AI recommendation system to personalise book and learning path suggestions based on the user's academic background. Users are also able to update their program selection at any time through their profile page.

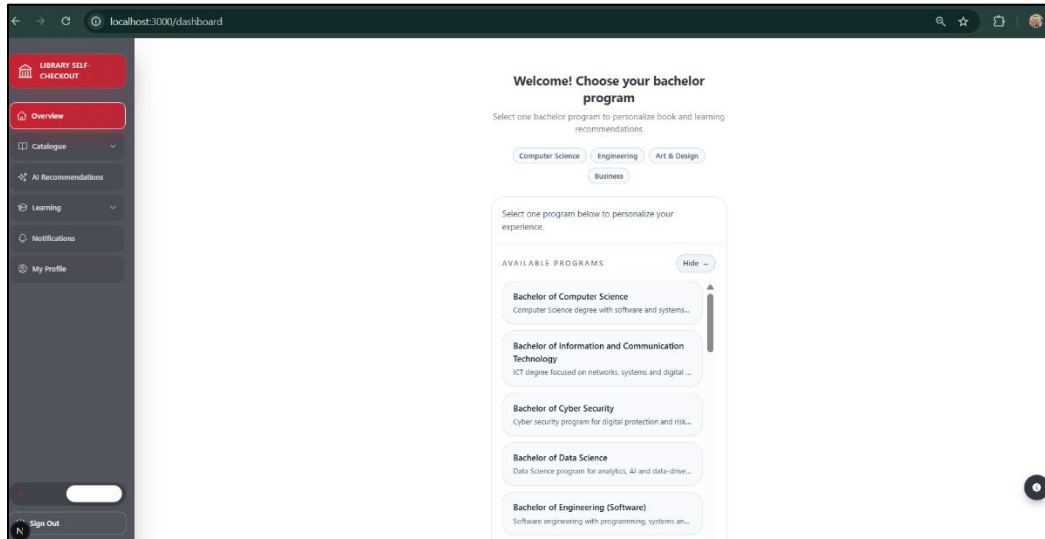
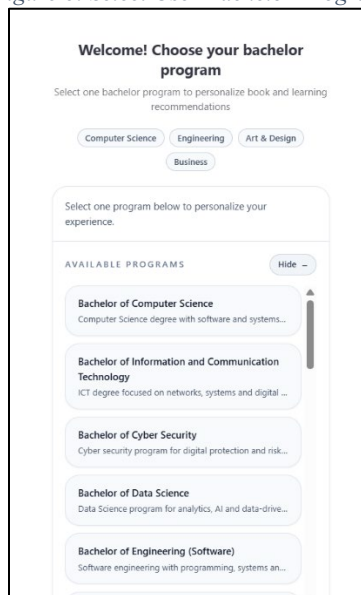


Figure 6: Select User Bachelor Program



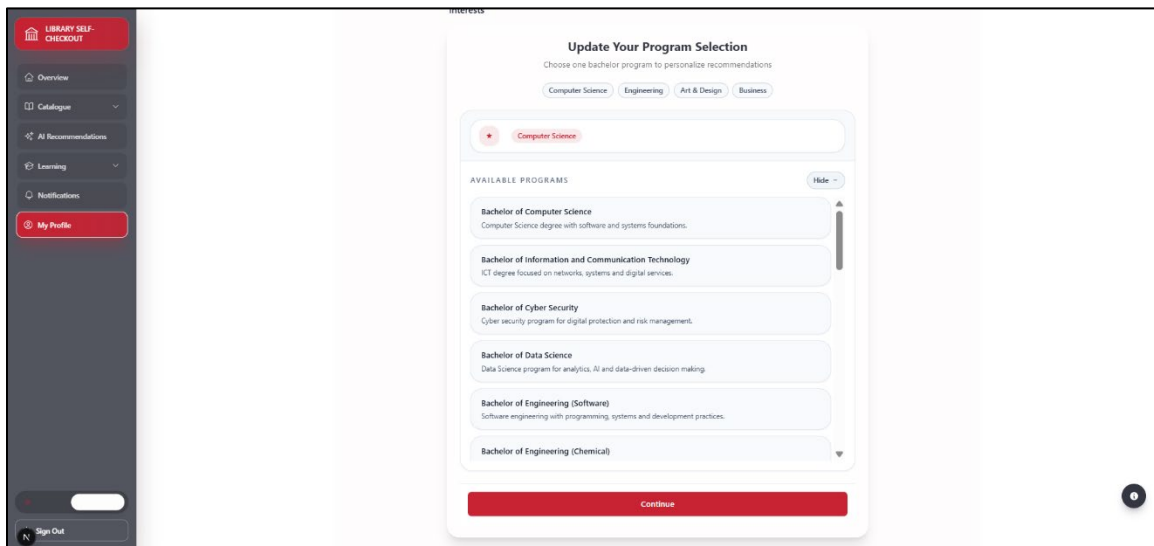


Figure 7: User Profile (Update Bachelor Program)

GitHub Contribution:



Kelvin Weng Kiong FONG

During this sprint, I contributed 24 commits with 10,565 line additions and 5,315 deletions, ranking second among all contributors for this sprint.

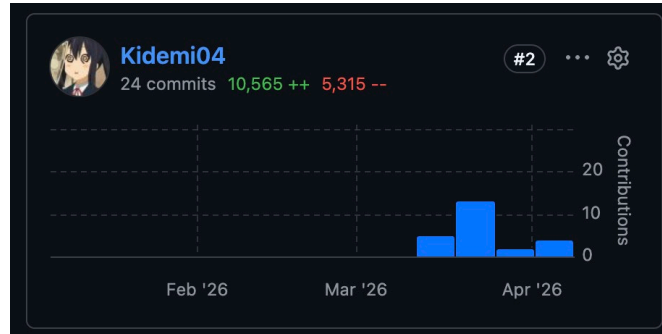


Figure 8: Sprint 2 GitHub contribution stats

To safeguard the team's stable progress before major changes were merged in, I created a dedicated backup branch from the main branch at a documented checkpoint, ensuring the team had a recoverable state during the high-risk database migration period.

Branches	
Overview	Yours
Active	Stale
All	
2026	
Branch	Updated
backip/main-2026-04-09	yesterday
backup/main-2026-03-27	2 weeks ago
main-backup-20260319	5 months ago

Figure 9: Backup branch with timestamp

I designed and implemented a complete borrowing history page for student users, which displays each loan with the book cover, title, author, ISBN, status, borrow and return dates, and calculated loan duration. The page also includes three summary stat cards at the top and a search and period filter bar. Access is restricted to student roles only.

I also wrote three automated test files for the project's Jest testing suite, covering the notification filter, sign out button, and tab switch components. These tests verify that the components render correctly and behave as expected when users interact with them.

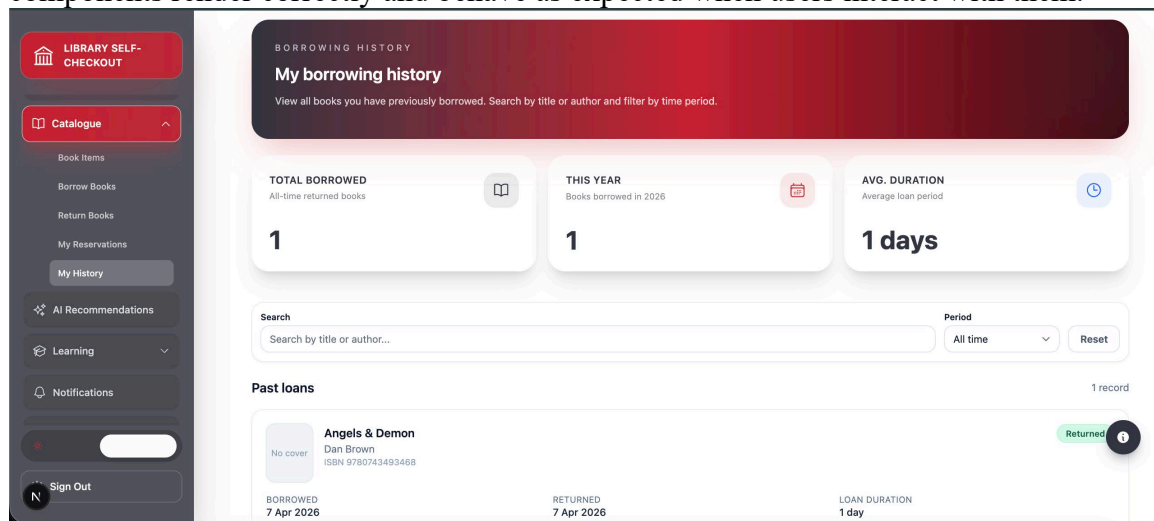


Figure 10: Borrowing history page UI

I also removed the entire Friends and Community social system from the codebase this sprint. After the team discussed it, we agreed that library users would unlikely use social features like following friends or joining communities, so keeping them in the system was adding unnecessary complexity. The removal involved deleting 21 files and over 2,250 lines of code, including the social pages, database query files, and related UI components. Throughout the sprint I handled all branch merges and conflict resolution for my teammates' pull requests. This sprint was particularly challenging for merges because the database migration Ivan was working on touched almost every part of the codebase, meaning most branches had conflicts that needed to be carefully resolved before merging to avoid breaking existing features. In total I merged four branches into main, covering Nigel's notification and holds system, Ivan's Supabase formatting update, Kenneth's LinkedIn feature, and my own RAG fix branch.

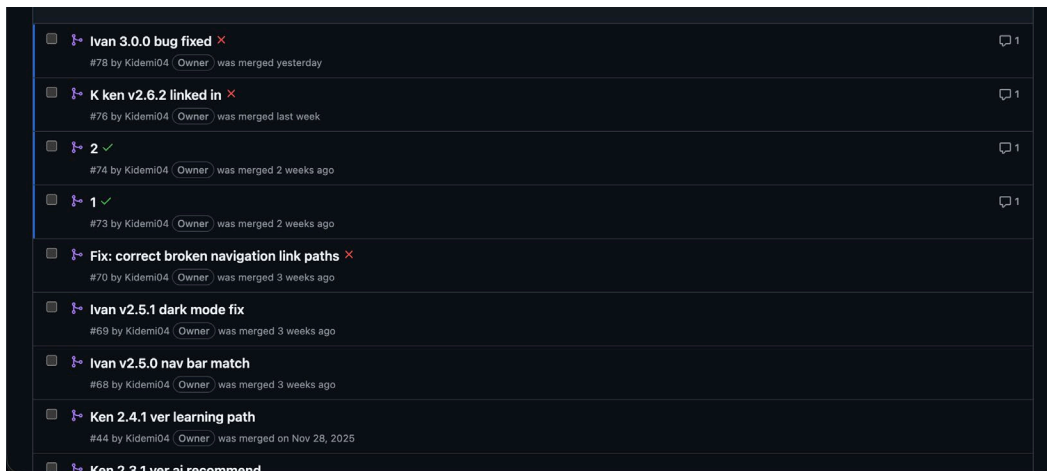


Figure 11: Merge history on GitHub

I also resolved a critical issue with the mobile barcode scanner, which was taking over 20 seconds to detect a barcode, making it practically unusable in a real library setting. After investigating, I found that the previous implementation was using a WebAssembly-based library called `zxing-wasm`, which struggled to process live camera frames reliably on mobile devices. I replaced it with a JavaScript-native library called `@zxing/browser`, which works directly with the browser without needing to load a heavy WebAssembly binary. The scanner now attempts to use the phone's built-in barcode detection first for speed, and only falls back to the library if that is not available. This brought the scan time down from over 20 seconds to under 2 seconds. I also fixed a database connection issue affecting the RAG-based AI chatbot built by a teammate.

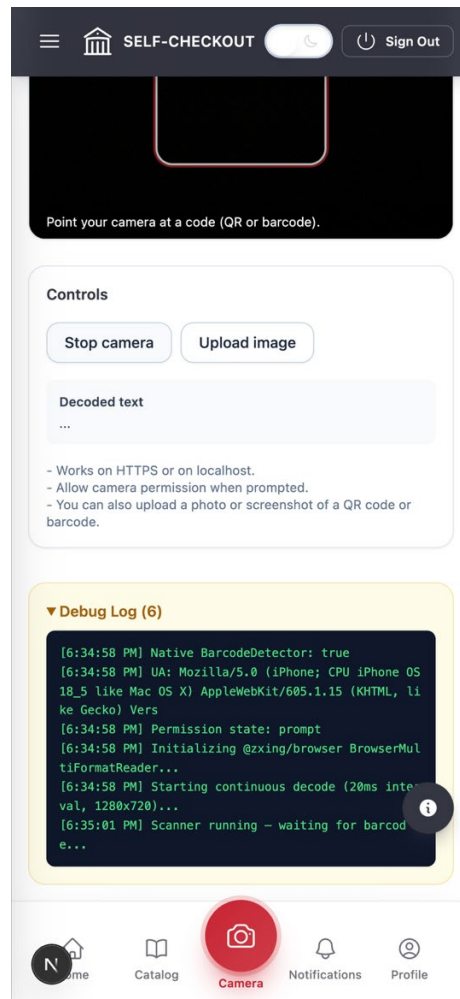
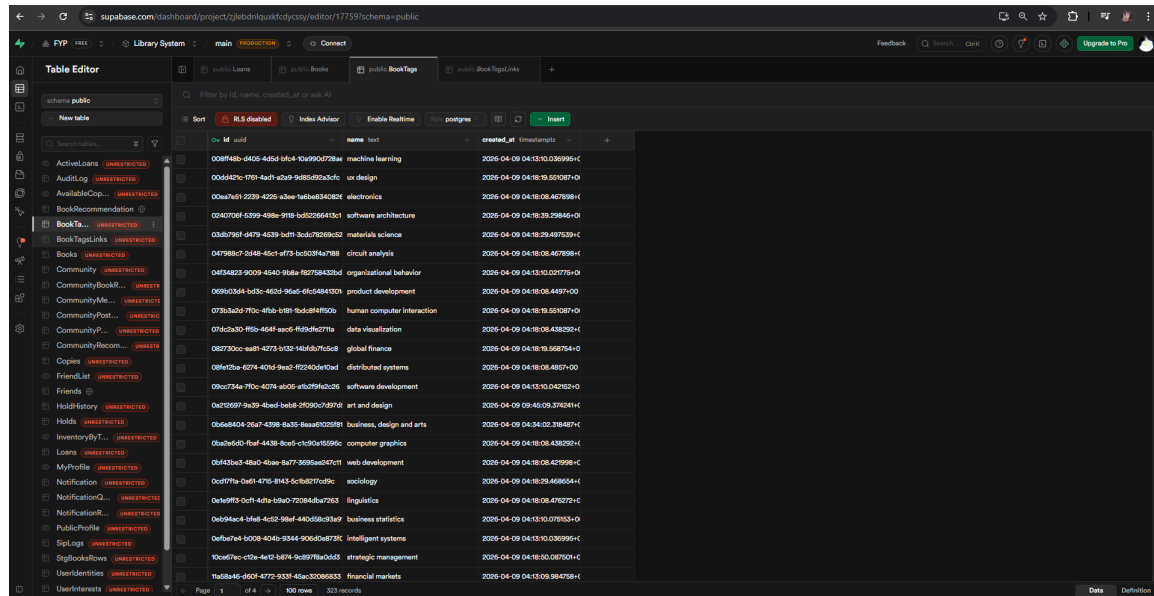


Figure 12: Mobile scanner UI

Nigel Zi Jun LING

In this sprint, I have helped the team to develop notification systems, refining reservation systems, generated every book a book tag, adding category search in catalogue page, and helped to implement learning path system and integrate it into AI recommendation system.

I first used Gemini AI to auto-tagged all the books, but then I realized the books all have different tags, and it's going to use all of the free tier usage of the AI model to retagged them all, so I decided to use AI to categorize them based on the tags. Right now, the categorize features are still not good enough so I added a manually categorized and tagged button for admin to manually edit those books to categorize them more precisely.



id	name	created_at	postgre
008f44b-4d05-4d5d-bf4d-10a90a702ae	machine learning	2026-04-09 04:13:10.36996+0	
00a4d2b-1701-4a4f-eb49-9485d92a3bf6	ux design	2026-04-09 04:18:19.55107+0	
00ea7d9-2239-4235-a3ee-1a6ba6343d0e	electronics	2026-04-09 04:18:08.407698+0	
024070f-5399-498a-9189-3d5226413c1	software architecture	2026-04-09 04:19:33.29846+0	
03da79f-0479-4639-b0ff-3dc78269d5d	materials science	2026-04-09 04:18:29.497639+0	
047989c-2448-45c1-e772-bc529f4a7f88	circuit analysis	2026-04-09 04:18:08.407698+0	
04f54823-9009-4540-9d8a-83756432bd	organizational behavior	2026-04-09 04:13:10.020775+0	
069a354-b43c-463d-96a5-6f6548430d	product development	2026-04-09 04:18:08.4487+0	
073b3a3d-710c-4f8a-b181-bd48f4ff0b	human computer interaction	2026-04-09 04:18:19.55107+0	
07d3da3d-7f0b-464f-aac0-f8d8f8c77f1a	data visualization	2026-04-09 04:18:08.438292+0	
082730c-ea81-4273-5132-1a6a6b7f5d8	global finance	2026-04-09 04:18:19.568754+0	
08f42ba-6274-407d-9a0c-f02a0a0c0ad	distributed systems	2026-04-09 04:18:08.4857+0	
09c734a-710c-407d-a005-a1027f6d0c5	software development	2026-04-09 04:13:10.042163+0	
0a2709f-3a39-4ba0-ba0b-3f090c7d97d	art and design	2026-04-09 09:45:09.374241+0	
0ba68404-26a7-4288-8a39-8a0a020f8f	business, design and arts	2026-04-09 04:34:02.318487+0	
0ba6a6d0-fa0f-4438-8a39-c1c50a10396c	computer graphics	2026-04-09 04:18:08.438292+0	
0d43ba3d-48a0-4bae-8a77-3659a0a03d11	web development	2026-04-09 04:18:08.421998+0	
0d077fa-0a61-471c-81a3-5c1a0271c99c	sociology	2026-04-09 04:18:29.468654+0	
0d49f93-0a7f-4d4a-b0a0-720848a7263	linguistics	2026-04-09 04:18:08.476722+0	
0e0d4ac4-bf4d-4c32-98af-440d8a0c3a9	business statistics	2026-04-09 04:13:10.07953+0	
0ef8a7e4-6008-404b-9344-906a0a0739c	intelligent systems	2026-04-09 04:13:10.36996+0	
10a07ec-c1a0-4a10-8a7a-9a9778a0a63	strategic management	2026-04-09 04:18:10.087501+0	
1a0b4a6-6024-4772-933f-45a02096533	financial markets	2026-04-09 04:13:09.964758+0	

Figure 13: BookTags table

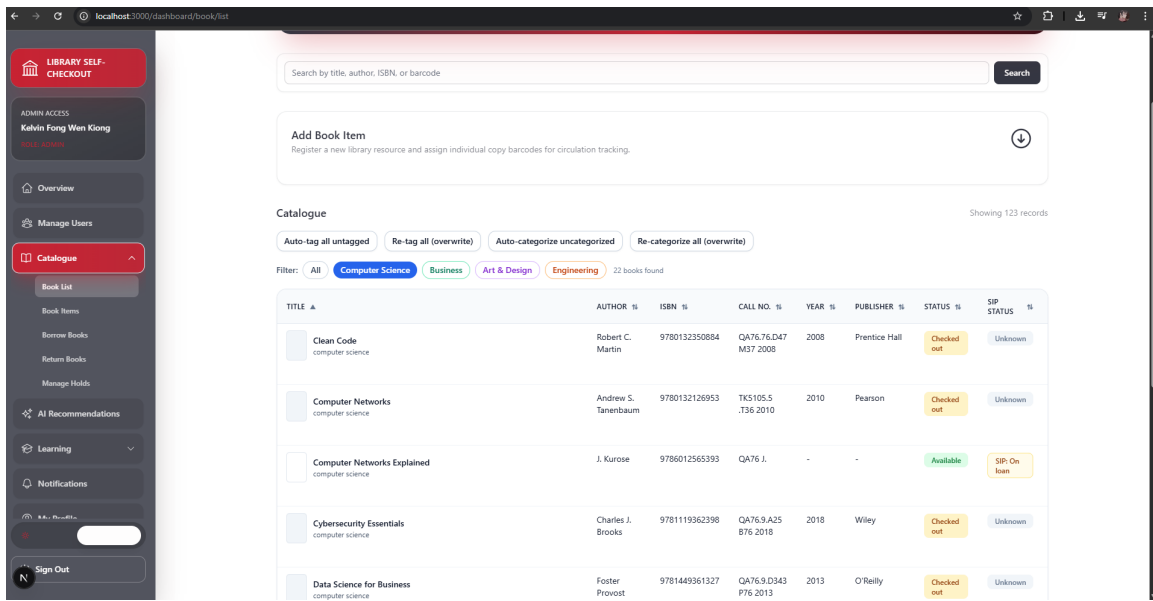


Figure 14: Category Search

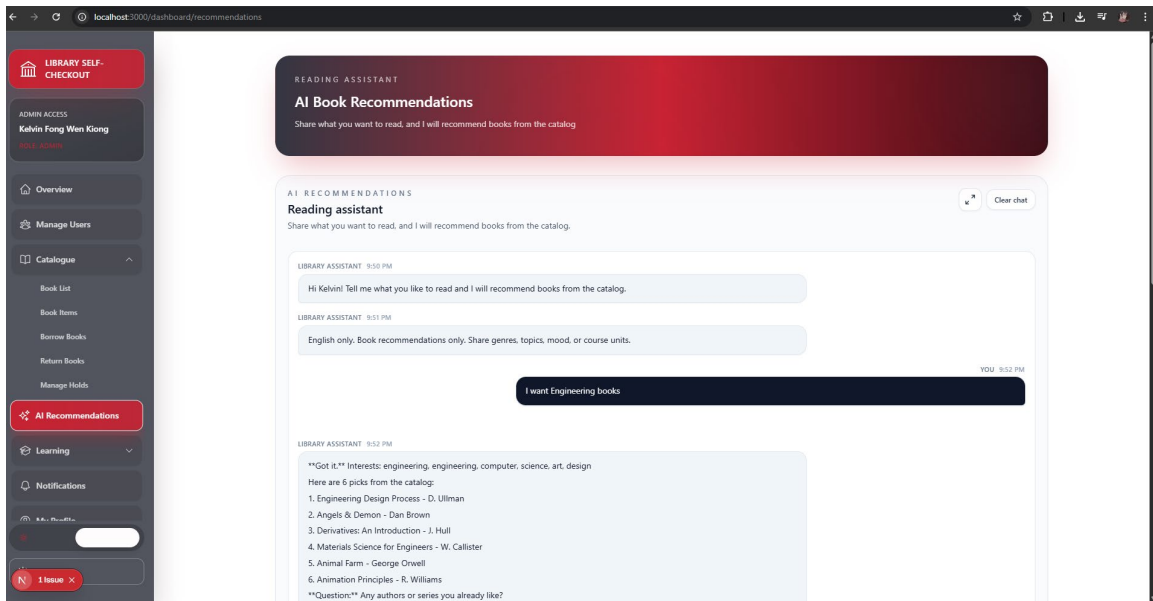


Figure 15: AI Book recommendation system

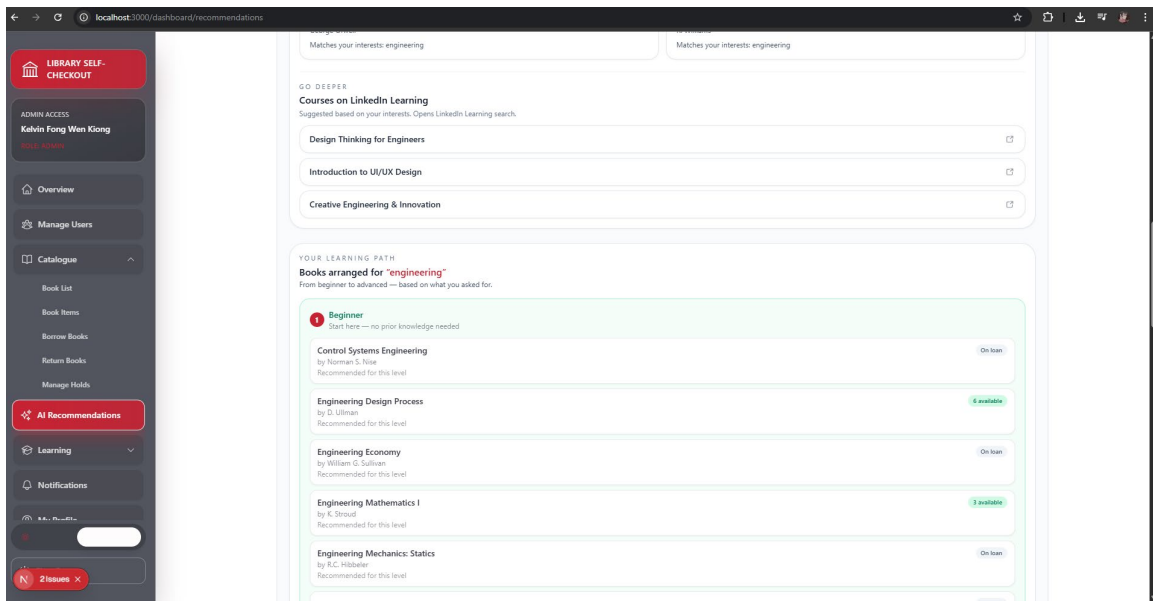


Figure 16: Learning Path system Beginner level

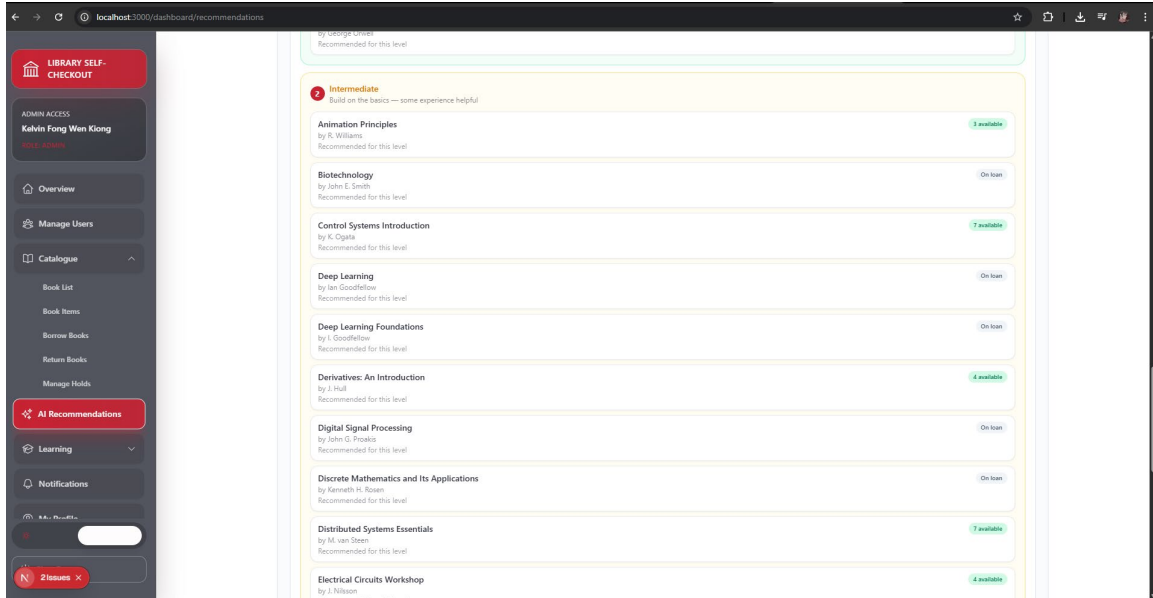


Figure 17: Learning Path system intermediate level

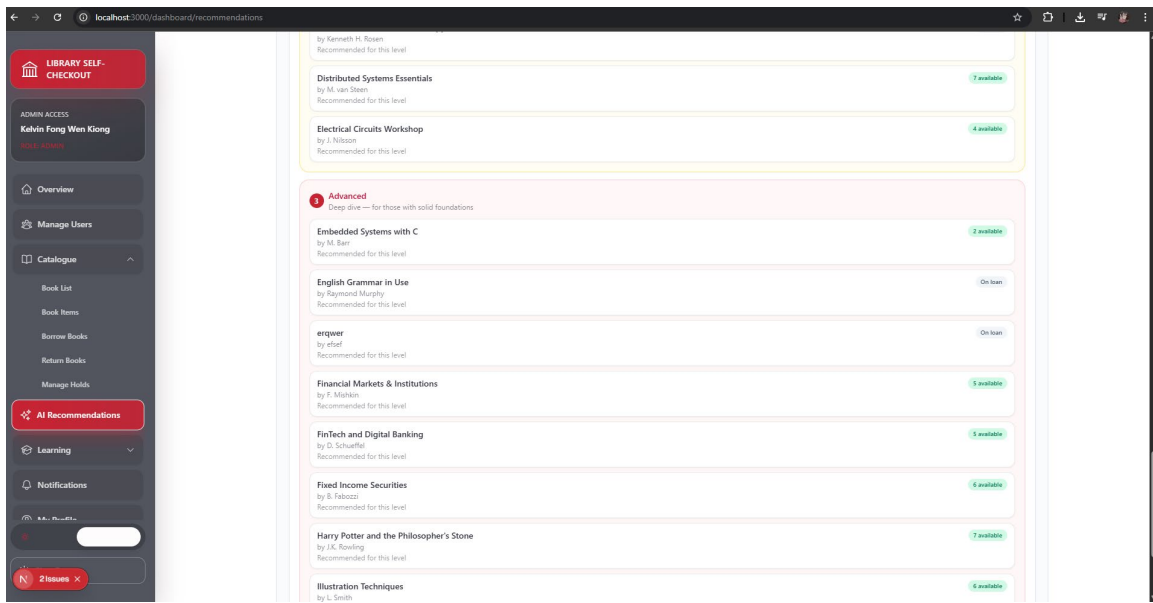


Figure 18: Learning path system advanced level

The AI recommendations are still not good enough to detect the important keywords when the user prompt is complicated, but right now it can work together with learning path system, the learning path systems will build a list based on the prompt.

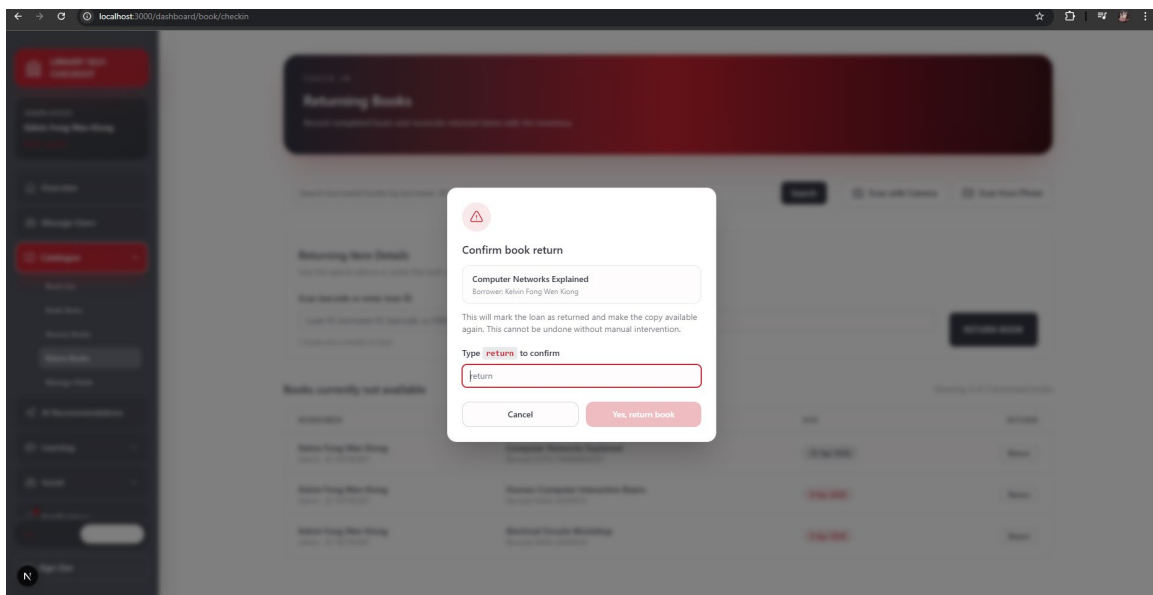


Figure 19: Confirm window for returning books

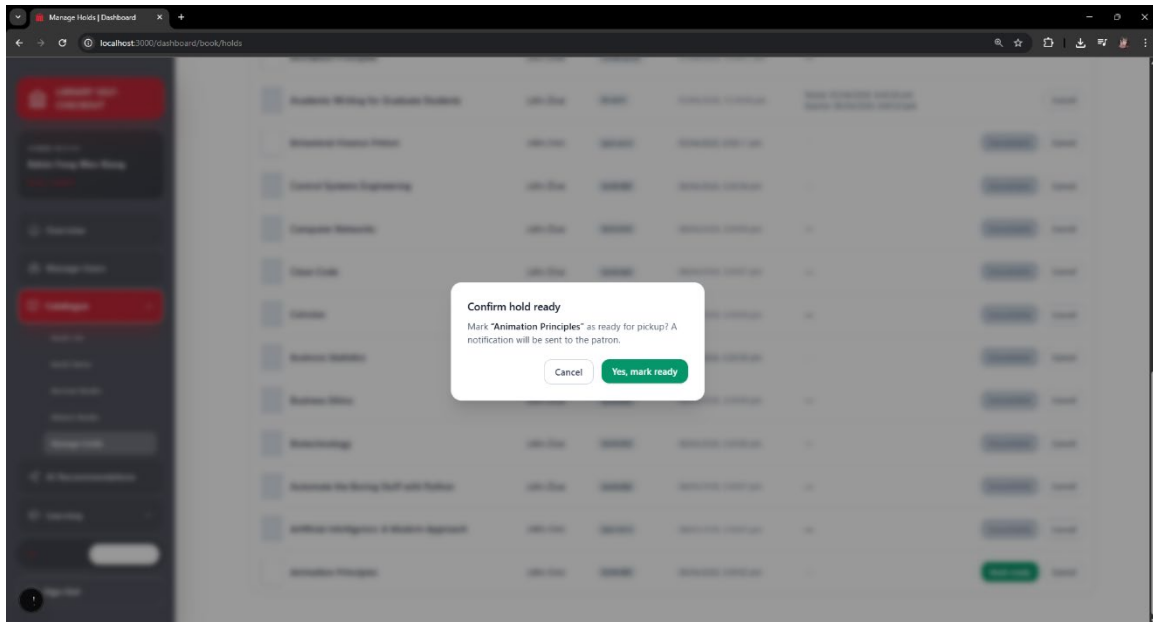


Figure 20: Confirm window of mark ready a book

I also added these 2 confirmation windows for return books page and manage holds page for admin and staff, now when they want to return a book, they need to manually type “return” to confirm return the book. And in “Manage Holds” page, admin/staff will get a double confirmation window to prevent accident click.

Ivan Jia Wei HOAN

As for this sprint, I am accountable for standardizing the format as well as optimizing the notation. As per the client’s requirement, filenames should follow the camelCase convention, the table names should adhere to the PascalCase convention, and the columns should follow the snake_case convention. In order to maintain the consistency, I have established a new Supabase format and shifted from the previous Supabase format to this new format. All format rules have been standardized and meet the client’s requirements. Apart from my contributions during this sprint, I actively assisted other people with their work as well, and this included activities like sending notifications and user engagement.

Database Tables

schema public 🔍 Search for a table + New table

NAME	COLUMNS	ROWS (ESTIMATED)	SIZE (ESTIMATED)	REALTIME	
ActiveLoans	10	–	–	✕ Disabled	View columns ⋮
AuditLog	11	88	72 kB	✕ Disabled	View columns ⋮
AvailableCopies	6	–	–	✕ Disabled	View columns ⋮
BookRecommendation	7	0	16 kB	✕ Disabled	View columns ⋮
Books	13	123	96 kB	✕ Disabled	View columns ⋮
BookTags	3	356	160 kB	✕ Disabled	View columns ⋮
BookTagsLinks	2	123	176 kB	✕ Disabled	View columns ⋮
Community	10	4	32 kB	✕ Disabled	View columns ⋮

Figure 21: Database Tables

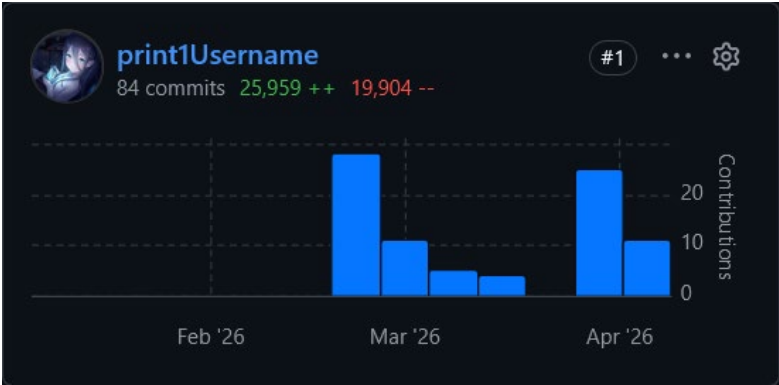


Figure 22: GitHub Commit (Last 3 Month)

1 Sprint Plan

1.1 SPRINT GOAL

In Sprint 2, the team decides to complete the implementation of the system and deploy online. We first target to move our entire system backend to the new database that matches the client's requirement format and fine-tuning or fix those functions so the system can work perfectly with the new database. We aim to develop the AI recommendation system by combining it with Learning Path integration and online courses from other organizations such as LinkedIn. Also, we want to make the AI system work as smartly as possible to provide useful and logical suggestions and detect the correct and needed keywords from user prompt. We also planned to optimize the UI to make it more user-friendly for new users. In the next sprint, we are planning to create google forms with detailed questions to test user acceptance, especially from librarians, collecting their reviews to do further improvements.

1.2 SCOPE

The scope of Sprint 2 is fully implemented the entire system of Library Self-Check Out websites, including performing basic features like check-in and check-out features, notifications system, manage user, reservation system, catalogue view, user management, friends system, also with enhancing features such as learning path system, AI-driven book recommendation system. The system implementation is finally completed and planned to start user testing in the next sprint.

1.3 ASSUMPTION & DEPENDENCIES

To ensure the successful delivery of Sprint 2, we are proceeding based on the following key assumptions and have identified primary dependencies:

1.3.1 Assumption 1: LinkedIn Learning Integration

We assumed that the LinkedIn Learning integration requested by the client would be technically feasible within the sprint. This assumption proved incorrect as LinkedIn Learning requires an institutional license and API access that is not available to us. Attempts to clarify the requirement with the client were also unsuccessful as no response was received. As a result, the LinkedIn Learning integration was dropped from the current sprint scope and will require further discussion with the client in Sprint 3.

1.3.2 Assumption 2: Gemini API Free Tier

We assumed that the free tier of the Gemini API would be sufficient to support the AI recommendation and RAG chatbot features throughout the sprint. In practice, the rate limits on the free tier were hit frequently, causing the API to become unresponsive during testing and development. This slowed down progress on AI related features and remains an ongoing constraint going into Sprint 3.

1.3.3 Assumption 3: Database Migration Scope

We assumed that switching the Supabase database to match the client's required naming convention would be a contained and manageable task. In reality, the naming changes affected queries across the entire codebase, requiring extensive debugging to identify and fix every affected reference.

1.3.4 Dependency 1: Gemini API

The Gemini API is used as the AI model powering both the book recommendation system and the RAG chatbot feature.

1.3.5 Dependency 2: Supabase

Supabase serves as the backend database provider that hosts all application data for the system.

1.3.6 Dependency 3: Vercel

Vercel is used as the hosting and deployment platform, with automatic deployments triggered on every push to the main branch.

1.3.7 Dependency 4: GitHub

GitHub serves as the version control system and CI/CD pipeline, and is also the primary tool for team progress synchronisation through pull requests and branch management.

1.4 CAPACITY & FORECAST

Team Member	Estimated Hours
Kenneth Hui Hong CHUA	20-25 hours
Ivan Jia Wei HOAN	20-25 hours
Kelvin Wen Kiong FONG	20-25 hours
Nigel Zi Jun LING	20-25 hours
Total	Around 120 hours

The estimated hours for each team member were determined based on the complexity of the tasks identified in the sprint backlog. Each member was allocated 20 to 25 hours, bringing the total team capacity to approximately 120 hours for the sprint. The majority of these hours were spent on AI related features including the automated book tagging system, the AI recommendation system, and the learning path integration, as these tasks required significant research, testing and iteration. A considerable portion of time was also spent on the database migration, which involved converting all table and column names to match the client's required naming convention and updating every affected query across the codebase. The remaining hours were distributed across bug fixing, feature implementation, and branch integration work. It is worth noting that team capacity was partially affected by other academic commitments during this period, which is reflected in the estimated rather than recorded nature of these figures.

1.5 DEFINITION OF DONE

A story is considered done when the system is fully developed with every feature and hosted online. For the base system, it should be performing well on all the functions that will interact with the backend such as check-in/out, notification, place holds after switching the backend database. We should create at least three user accounts for testing the AI recommendation system, each user should simulate the real scenario cases by borrowing certain types of books to provide history data to read by the AI to do recommendation. For the AI recommendations system, it should be able to recommend books based on user's borrowed books history, followed by suggested learning path that generated lists of books of different difficulty for user to study. To achieve this, the system should tag every book in the database and use AI to categorize it into four categories, which are Computer Science, Engineering, Art & Design, and Business. Furthermore, the websites UI should be more convenient and user-friendly, especially for the first time user.

1.6 SPRINT MILESTONE & REVIEW ARTIFACTS

During these 3 weeks of Sprint 2, our main target is to finish the system implementation and make the system as practical as possible. We want to finish every possible in this sprint so we can prepare for our user acceptance test in the next phase. Furthermore, these sections also included the evidence of the progress we have achieved, presented by GitHub commit history, screenshots from the websites, also with test logs from backend testing. Our team has at least one weekly online meeting through Microsoft Teams and updating latest progress in WhatsApp group and also having physical meetings for almost everyday to keep track of the current plan.

1.7 SPRINT BACKLOG

ID	Item	Acceptance criteria	Est.hrs.	Assignee
S2-01	Notification system	Users should be able to receive notification of check-in/out, put holds, reserved book available, book return due date warning, AND admin/staff should receive check-in/out, holds place -All roles should be able to view notification in “Notification” page -Unread message will show as a pop-up small window.	10	Nigel
S2-02	Manage Holds & Reservations	Only first user in the queue should be able to receive notification of available, Admin & staff can only mark a book ready when the book is returned.	5	Nigel
S3-03	Manage Supabase Database	The Supabase table format should reach the client requirements, Database Table name with PascalCase, Database table column name with snake_case	20	Ivan
S3-04	Create User Interest	Create user interest for AI book recommendation	5	Ivan
S3-05	Create Book tags	All book should have a book tag in database	5	Nigel
S3-06	Create Filter in Catalogue	All books in “Book List” page should be categorized into Four category (Computer Science, Business, Art & Design, Engineering)	5	Nigel
S3-07	AI recommendations combine with Learning path system	Now the AI recommendation chatbot will together combine with learning path system	5	Nigel
S3-08	Barcode Scanner Fix	Scan time should be under 3 seconds on mobile devices	5	Kelvin
S3-09	Borrowing History Page	Student users can view past loans with search and filter functionality	5	Kelvin

S3-10	Branch Integration & Merge Management	All teammate branches merged into main without breaking existing features	3	Kelvin
S3-11	Backup Branch	Stable backup branch created from main before major changes	1	Kelvin
S3-12	AI Recommendation System Improvement	AI chatbot accurately recommends books based on user interest and borrowing history	8	Kenneth, Nigel, Kelvin
S3-13	LinkedIn Learning Integration	Users can search and browse LinkedIn Learning courses from the dashboard	10	Kenneth
S3-14	AI Guardrails & Content Safety	AI only responds to library related queries and rejects off-topic inputs	3	Kenneth
S3-15	User Interest based on Bachelor Program	AI recommendation considers user's academic program when suggesting books	5	Kenneth

2 Quality management plan and outcome

At this stage, we have the majority of the solution developed, it is a good time to finalize the quality management plan. This section should include the details of any quality management activities undertaken already and all that are planned. Please note that a good quality management plan should not only state the goal, but also detail how to assess whether the goal has been met or not. This can be an incomplete work; however, please include the results of the testing that has been done already and the tests that are planned for the future.

2.1 QUALITY GOALS (WHAT YOU AIM FOR)

The team set two main quality goals for this sprint. First, the system should perform smoothly on mobile devices, maintaining responsive layouts and fluid animations without noticeable lag or performance issues. This is important because the system is intended to be used by students on their phones for self checkout. Second, all features should be stable enough to pass a successful build and deployment on Vercel, as any broken code automatically blocks the deployment from going live and affects the entire team.

2.2 QUALITY MANAGEMENT ACTIVITIES (WHAT YOU'VE DONE SO FAR)

In this sprint, the team carried out three types of quality management activities to ensure the system remained stable and functional throughout development.

On the automated testing side, the project uses Jest and React Testing Library, both integrated with the Next.js build environment. Ivan and Kelvin wrote six test files covering 28 individual test cases, targeting core user interface components including the homepage, login page, navigation tabs, sign out button, notification filter, and a UI button component. Whenever changes are made to the codebase, these tests run automatically to verify that the components still render correctly and behave as expected.

The team also carried out manual testing together throughout the database migration process. Everyone participated in checking that key features such as login, logout, book checkout, book checkin, and general navigation still worked correctly after the naming convention changes were applied. Testing was done on an iPhone 15 Pro and a MacBook to make sure the system held up on both mobile and desktop.

On top of that, every push to the main branch automatically triggers a build check on GitHub Actions followed by a deployment on Vercel. If any code breaks the build, the deployment is blocked straight away, which keeps the live system clean and stable at all times.

2.3 TESTING RESULTS (EVIDENCE OF QUALITY)

The automated Jest test suite was run and all 28 tests across 6 test suites passed successfully, completing in just over 2 seconds.

```
(base) kelvinfong@Kelvins-Macbook Library_Self-Checkout_System % npm test
at app/ui/dashboard/signOutButton.tsx:22:27

console.error
  Error: Not implemented: navigation (except hash changes)
    at <unknown> (app/ui/dashboard/signOutButton.tsx:22:27) {
      type: 'not implemented'
    }

20 |         await signOut({ redirect: false });
21 |         // Then end the Azure AD SSO session so the next sign-in sh
> 22 |         window.location.href = '/api/auth/azure-signout';
    |                                ^
23 |     });
24 | };
25 |

at VirtualConsole.<anonymous> (node_modules/.pnpm/@jest+environment-js
environment-jsdom-abstract/build/index.js:87:23)
at module.exports (node_modules/.pnpm/jsdom@26.1.0/node_modules/jsdom/
at navigateFetch (node_modules/.pnpm/jsdom@26.1.0/node_modules/jsdom/
at exports.navigate (node_modules/.pnpm/jsdom@26.1.0/node_modules/jsdom
at LocationImpl._locationObjectNavigate (node_modules/.pnpm/jsdom@26.1
on-impl.js:30:5)
at LocationImpl._locationObjectSetterNavigate (node_modules/.pnpm/jsdom
Location-impl.js:24:17)
at LocationImpl.set href [as href] (node_modules/.pnpm/jsdom@26.1.0/no
pl.js:46:10)
at Location.set href [as href] (node_modules/.pnpm/jsdom@26.1.0/node_m
125:37)
at app/ui/dashboard/signOutButton.tsx:22:27

PASS  __tests__/homePage.test.jsx
PASS  __tests__/notificationFilter.test.jsx

Test Suites: 6 passed, 6 total
Tests:      28 passed, 28 total
Snapshots:  0 total
Time:       2.206 s
Ran all test suites.
(base) kelvinfong@Kelvins-Macbook Library_Self-Checkout_System %
```

Figure 23: Jest test results showing 28 passed, 6 suites

Manual testing during the database migration did not reveal any major bugs. All key features continued to function correctly after the naming convention changes were applied across the codebase. And Vercel deployment builds remained stable throughout the sprint with no major build failures blocking progress.

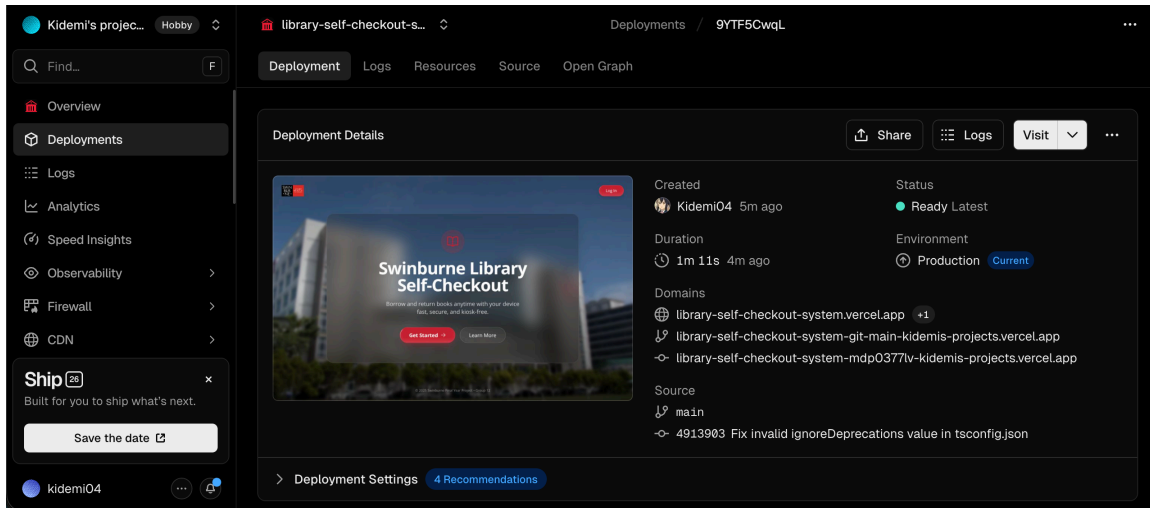


Figure 24: Vercel build checks

2.4 EVALUATION (DID YOU MEET THE GOALS?)

The deployment stability goal was fully met. Every build passed successfully and the live system remained accessible throughout the sprint. The mobile performance goal was partially met as the system runs smoothly on the tested devices, however formal performance benchmarking such as load time measurement has not yet been conducted. Automated test coverage remains limited, with only surface level UI components covered and no tests yet written for the core API endpoints, database queries, or AI related features.

2.5 FUTURE QUALITY PLANS (WHAT YOU WILL IMPROVE/TEST NEXT)

Based on the evaluation above, the team has identified two main areas to address in Sprint 3. First, automated test coverage will be expanded to include the checkout and checkin API endpoints and the core database query functions, as these are the most critical parts of the system and currently have no test coverage. Second, user acceptance testing will be conducted where real users interact with the system to identify usability issues that automated tests cannot catch. Manual testing will continue to be performed on both mobile and desktop devices as new features are introduced.

3 Sprint Progress

This section describes the progress of the sprint with relevant evidence (e.g., screenshots of the repository, task board, etc). The evidence can be discussed with the supervisor.

3.1 SPRINT DELIVERABLES

During these 3 weeks of Sprint 2 process, we tried to finish the last few of remaining tasks in our system such as improving our AI-recommendation system, switching our backed database, UI refining, small features adding for convenience purpose and fixing other minor bugs.

During the development process, we do find that the LinkedIn system that is required by our client is impossible to implement into our system since it requires institution license and certificate to integrate with their online course into our system. After consulting with our unit convenor, Mr. Brian Loh, he suggested we switch to another similar online course that is free to put in our system. So, because of that, we are trying to find out alternative feature to replace LinkedIn system.

For the AI recommendations system, we decided to test it out by making a real case scenario. First, we are creating some students(user) accounts that only borrow certain types of books (such as Student A only borrow Engineering topic books) to let the AI to read through the user's history and make recommendations based on it.

3.1.1 User Interest

In order for the AI to have a better understanding of the users' preferences and be able to suggest books to them that will fit their needs, we have included a feature called 'User Interests' in the user's profile page.

The existing four categories are Business, Computer Science, Design & Engineering. The user can choose the type of book that is appealing to him/her based on his/her interests.

At the time of their first-ever log-in, new users have to pick their interests from among four choices. This choice screen will pop up on the very first log-in attempt and will never come back into play afterward.

Welcome! Choose your bachelor program

Select one bachelor program to personalize book and learning recommendations

Computer Science Engineering Art & Design Business

Select one program below to personalize your experience.

AVAILABLE PROGRAMS Hide -

Bachelor of Computer Science
Computer Science degree with software and systems fou...

Bachelor of Information and Communication Technology
ICT degree focused on networks, systems and digital ser...

Bachelor of Cyber Security
Cyber security program for digital protection and risk m...

Figure 25: Choose User Interest

Interests

Update Your Program Selection

Choose one bachelor program to personalize recommendations

Computer Science

Engineering

Art & Design

Business

★

Bachelor of Computer Science

Computer Science degree with software and systems foundations.

Computer Science

AVAILABLE PROGRAMS

Hide -

Bachelor of Computer Science

Computer Science degree with software and systems foundations.

Bachelor of Information and Communication Technology

ICT degree focused on networks, systems and digital services.

Bachelor of Cyber Security

Cyber security program for digital protection and risk management.

Bachelor of Data Science

Figure 26: Update User Interest

Users

UserInterests

Filter by user_id, interests, updated_at or ask AI

Sort

RLS disabled

Index Advisor

Enable Realtime

Role postgres

Insert

	user_id uuid	interests text[]	updated_at timestampz	
	e0452a86-1ce7-40b8-a1b2-80295...	["bachelor_of_business_finance_internati	2026-04-09 14:55:23.77+00	
	f8f236a9-d995-4403-9344-9ee7f...	["bachelor_of_computer_science"]	2026-04-09 15:14:27.677+00	

Figure 27: User Interest Table

UNIT CODE COS40006

SPRINT 2 REPORT

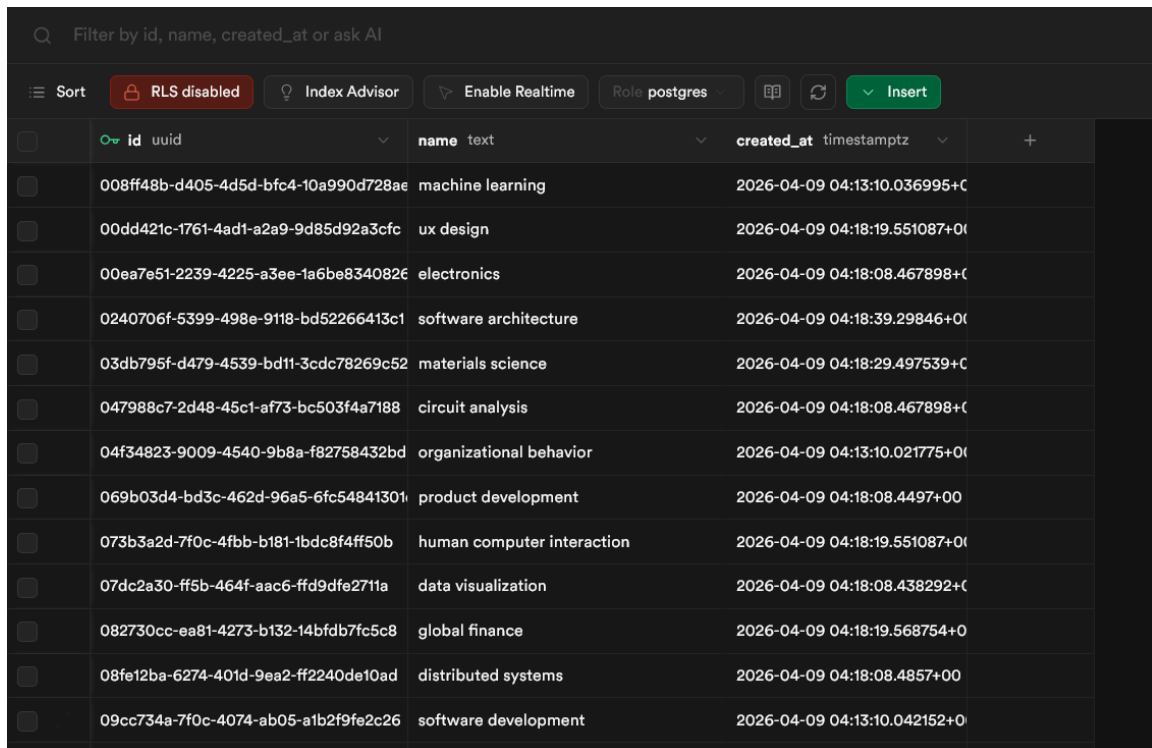
PAGE 28

3.1.2 Add Book Tags

To classify the books easily and help the AI provide more appropriate books to the users, we have introduced tags for the books. The users can look up based on these tags, and even the AI can suggest books to the users based on these tags.

We have also classified these books in four different classes: Business, Computer Science, Design & Engineering, and saved in the database so that they can be searched using these classes. On the other hand, through these classes, AI can find those books which suit the user.

And now the user can search the books in catalogue based on the categories.



id	name	created_at
008ff48b-d405-4d5d-bfc4-10a990d728ae	machine learning	2026-04-09 04:13:10.036995+00
00dd421c-1761-4ad1-a2a9-9d85d92a3cfc	ux design	2026-04-09 04:18:19.551087+00
00ea7e51-2239-4225-a3ee-1a6be8340826	electronics	2026-04-09 04:18:08.467898+00
0240706f-5399-498e-9118-bd52266413c1	software architecture	2026-04-09 04:18:39.29846+00
03db795f-d479-4539-bd11-3cdc78269c52	materials science	2026-04-09 04:18:29.497539+00
047988c7-2d48-45c1-af73-bc503f4a7188	circuit analysis	2026-04-09 04:18:08.467898+00
04f34823-9009-4540-9b8a-f82758432bd	organizational behavior	2026-04-09 04:13:10.021775+00
069b03d4-bd3c-462d-96a5-6fc54841301	product development	2026-04-09 04:18:08.4497+00
073b3a2d-7f0c-4fbb-b181-1bdc8f4ff50b	human computer interaction	2026-04-09 04:18:19.551087+00
07dc2a30-ff5b-464f-aac6-ffd9dfe2711a	data visualization	2026-04-09 04:18:08.438292+00
082730cc-ea81-4273-b132-14bfdb7fc5c8	global finance	2026-04-09 04:18:19.568754+00
08fe12ba-6274-401d-9ea2-ff2240de10ad	distributed systems	2026-04-09 04:18:08.4857+00
09cc734a-7f0c-4074-ab05-a1b2f9fe2c26	software development	2026-04-09 04:13:10.042152+00

Figure 28: Book Tags Table

Filter by book_id, tag_id or ask AI

Sort RLS disabled Index Advisor Enable Realtime Role postgres

	book_id uuid	tag_id uuid
	05a415cc-f1d7-46cf-ace6-aa7fb2e...	0a212697-9a39-4bed-beb8-2f090...
	06222033-816c-459a-b756-3f590...	1466640b-b9f6-41e7-868f-1fda6f...
	0c1550e7-4697-4abf-a75d-58cc3...	95eede07-9598-47f5-ae38-544b8...
	0c79fb21-36d6-46dd-9cc7-64473...	95eede07-9598-47f5-ae38-544b8...
	0ecbbb5b-c941-4ec7-b9e1-a7214...	95eede07-9598-47f5-ae38-544b8...
	0f4b95fe-7323-4dc7-af80-e192d1...	95eede07-9598-47f5-ae38-544b8...
	0ff01170-d3e5-4e89-8e50-29760...	91ac3a16-dc76-42a1-85b8-801cc1...
	135e19ba-94a4-468d-bc9e-1cc155...	0a212697-9a39-4bed-beb8-2f090...
	166a694f-8555-432b-817d-fa6c88...	95eede07-9598-47f5-ae38-544b8...
	18b219d1-35f0-45e1-a5b9-c12866...	0a212697-9a39-4bed-beb8-2f090...
	1937d1fd-98b1-4779-b659-ab880...	95eede07-9598-47f5-ae38-544b8...
	1f90d630-4944-4a06-8189-1ef98...	91ac3a16-dc76-42a1-85b8-801cc1...
	22e52380-5252-4a07-867b-f11be6...	0a212697-9a39-4bed-beb8-2f090...

Figure 29: Book Tags Links Table

localhost:3000/dashboard/book/list

LIBRARY SELF-CHECKOUT

ADMIN ACCESS
Kevin Fong Wen Kiong

Overview
Manage Users
Catalogue
Book List
Book Items
Borrow Books
Return Books
Manage Holds
AI Recommendations
Learning
Notifications
Sign Out

Add Book Item
Register a new library resource and assign individual copy barcodes for circulation tracking.

Catalogue
Showing 123 records

Auto-tag all untagged Re-tag all (overwrite) Auto-categorize uncategorized Re-categorize all (overwrite)

Filter: All Computer Science Business Art & Design Engineering

TITLE	AUTHOR	ISBN	CALL NO.	YEAR	PUBLISHER	STATUS	SIP STATUS	ACTIONS
Academic Writing for Graduate Student engineering	John M. Swales	9780472034758	PE1410 .S33 2012	2012	University of Michigan Press	Checked out	Unknown	Manage Delete
Angels & Demon engineering	Dan Brown	9780743493468	THR BRO	-	-	Available	SIP: Available	Manage Delete
Animal Farm engineering	George Orwell	9780451526342	FIC ORW	-	-	Available	SIP: On loan	Manage Delete
Animation Principles engineering	R. Williams	9780910911863	N RL	-	-	Available	SIP: Available	Manage Delete
Artificial Intelligence: A Modern Approach art and design	Stuart Russell	9780134610993	Q335 .R86 2016	2016	Pearson	Checked out	Unknown	Manage Delete
Automate the Boring Stuff with Python art and design	Al Sweigart	9781593279929	QA76.73.P98 594 2019	2019	No Starch Press	Checked out	Unknown	Manage

Figure 30: Category Search

3.1.3 Database Switching

In order to meet the formatting needs of the client and to ensure better organisation, we have opted for creating a new Supabase database and transferring our data to the new database.

In accordance with the client's requirements, the names of all database tables must be converted to PascalCase, with the first letter of each word capitalised and the words joined together; at the same time, all column names must be converted to snake_case, with all letters in lower case and the words separated by underscores.

Since we have made changes in how the database is named, we must also update our code to suit that change. There are several areas within the code that talk about databases within the Supabase platform. However, when these changes are not made, there could be problems in finding the database. Thus, this makes it imperative for us to update our code.

As replacing a database outright is a major undertaking requiring the inspection of virtually all the code, this task is extremely time-consuming and labour-intensive. Furthermore, it is necessary to ensure that the new database retains the format of the old one and that the data within it matches that of the previous version; consequently, this task requires considerable time.

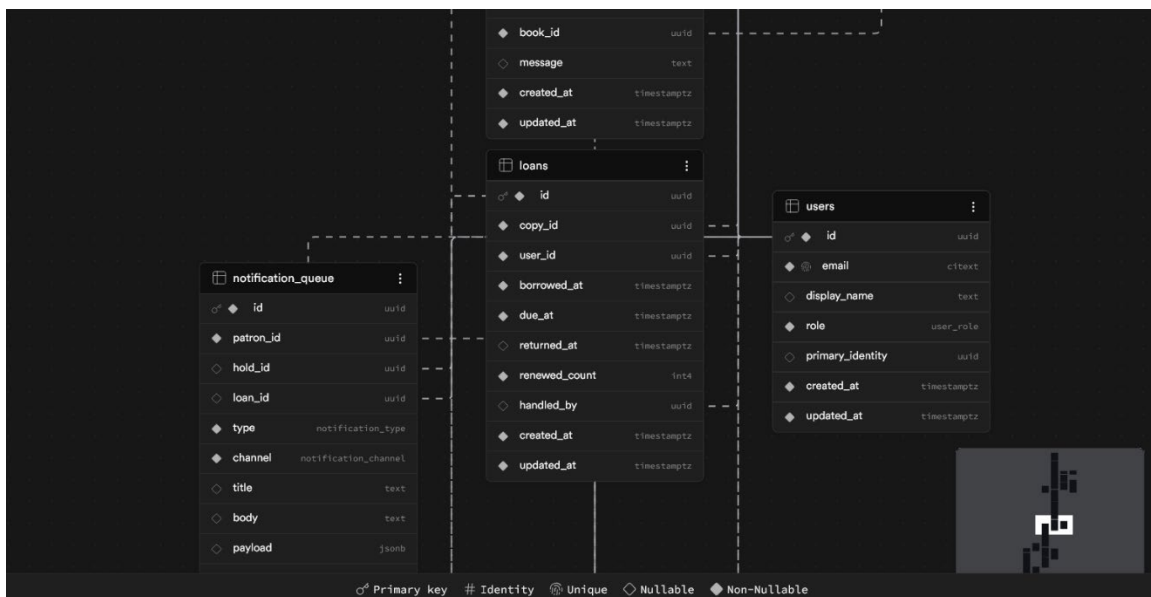


Figure 31: Old Database

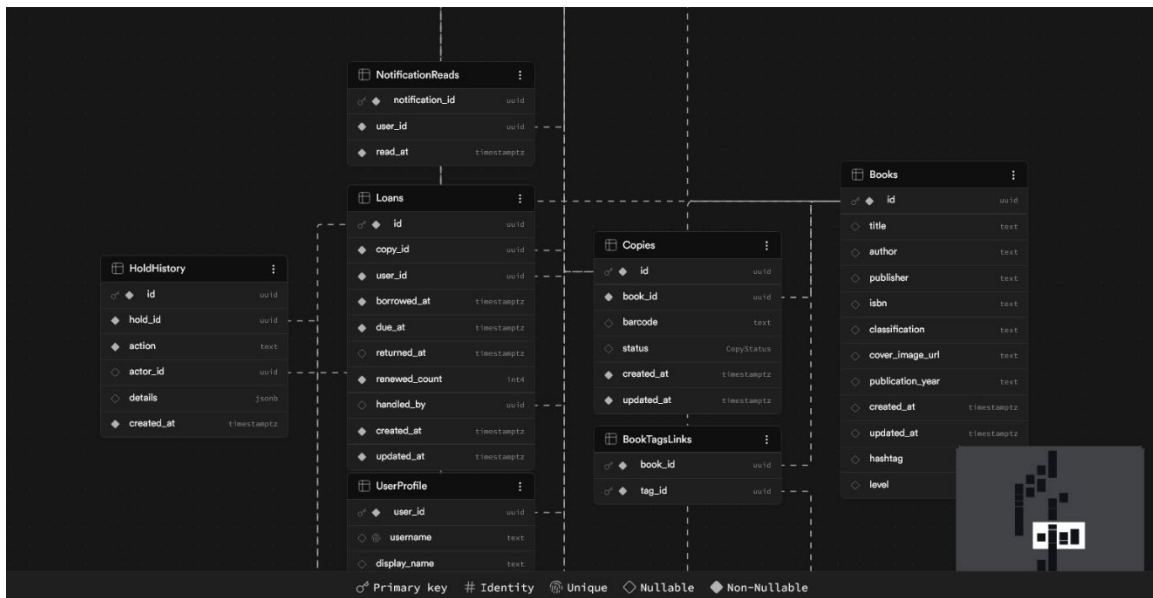


Figure 32: New Database

The screenshot shows a code editor with a commit history on the left and a diff view of a TypeScript file on the right. The commit history includes:

- typo000000
- Update updates.ts
- fix supabase
- Update .env.local
- Update slp2.ts
- Update updates.ts
- Update actions.ts
- check in
- Update actions.ts
- Update actions.ts
- fix borrow book
- Delete page.jsx
- edit manage holds page and rese...
- Update page.jsx

The diff view shows the following changes in `app/dashboard/actions.ts`:

```

@@ -883,7 +883,7 @@ export async function createBookAction(
 883   const copyRows = barcodes.map((barcode) => ({
 884     book_id: bookRow.id,
 885     barcode,
 886     status: 'available',
 887   }));
 888   const { error: copyInsertError } = await supabase.from('copies').insert(copyRows);
 889
 890   const copyRows = barcodes.map((barcode) => ({
 891     book_id: bookRow.id,
 892     barcode,
 893     status: 'available',
 894   }));
 895   const { error: copyInsertError } = await supabase.from('copies').insert(copyRows);
  
```

Figure 33: Table Update (Code)

3.1.4 Notification System

The notification system can be considered a very important element. When there is a notification system, everybody will get their notifications without delay, and this also increases efficiency.

If a user uses the book reservation service, then he or she is able to make reservations for books that he or she would like to read. The user will receive notification of placed holds. When such books are available, the user will be notified that he or she can borrow them.

As the due date for the return of a book draws near, the system will alert the user that the due date is nearing (3 days before the due date) and that the user must begin preparations to return the book.

On the other hand, for the purpose of preventing notification clutter due to excessive notifications, we are implementing a limitation that states that only the latest 50 notifications can be displayed, and the rest are not.

All the user will received notification of check-in/out. User will only receive notification of it

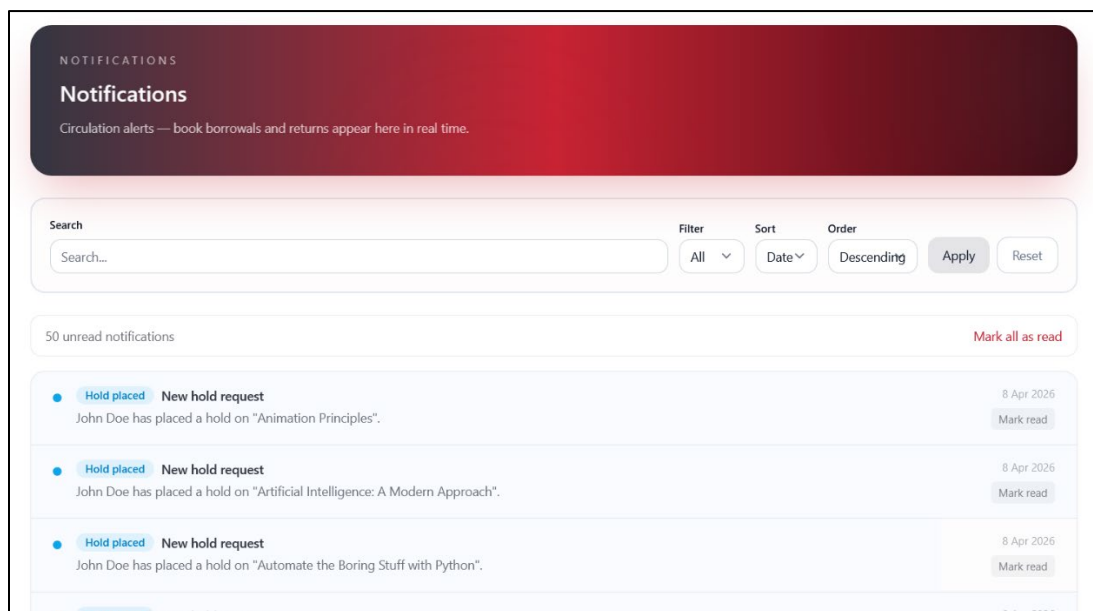


Figure 34: Notification Page

Filter by id, type, title... or ask AI

Sort RLS disabled Index Advisor Enable Realtime Role postgres Insert

	id	uuid	type	NotificationType	title	text	message	text	target_roles	jsonb
	07a6c564-8c63-4d18-800c-e50fe1954c2c		hold_ready		Book Borrowed		"Typography: Form and Function" was ch		["staff","admin"]	
	30bd4363-a5c7-4035-9d3d-af4c1404cef		hold_ready		Book Borrowed		"Human-Computer Interaction Basics" w		["staff","admin"]	
	3c7d3991-cad2-42bd-bcaa-5123776e23ca		hold_ready		Book Returned		"Control Systems Introduction" has been		["staff","admin"]	
	420c4cb9-81d0-4f99-bc44-c59acb56719f		hold_ready		Book Borrowed		"Financial Markets & Institutions" was che		["staff","admin"]	
	46438e40-8127-4832-9bf4-3bb0cbff35c9		hold_ready		Book Borrowed		"Design Thinking Workshop" was checke		["staff","admin"]	
	60912c88-3b94-4d3b-afd9-a2608815bc9f		hold_ready		Book Borrowed		"Computer Networks Explained" was che		["staff","admin"]	
	630bc018-02ef-4612-b421-5d99d58c3e31		hold_ready		Book Borrowed		"Deep Learning Foundations" was checke		["staff","admin"]	
	6e8e2977-3a2a-4a26-84aa-40622b9d7ac1		hold_ready		Book Returned		"Thinking, Fast and Slow" has been return		["staff","admin"]	
	8ced09ea-6362-49de-890a-6986d7ac75b		hold_ready		Book Returned		"Harry Potter and the Philosopher's Stone		["staff","admin"]	
	9e228e67-0705-434d-9900-d3c98c0004i		hold_ready		Book Returned		"Typography: Form and Function" has bee		["staff","admin"]	
	b515f2e9-fbf7-4bb3-ad20-da751ee24f2d		hold_ready		Book Borrowed		"Control Systems Introduction" was checke		["staff","admin"]	
	b69df71f-5310-4c20-8cb0-d00f245afc24		hold_ready		Book Borrowed		"User Experience Design 101" was checke		["staff","admin"]	
	c8cb9c14-e23c-4fc3-b75e-f8bd2a8718d0		hold_ready		Book Borrowed		"Computer Networks Explained" was che		["staff","admin"]	

Figure 35: Notification table

localhost:3000/dashboard/admin

LIBRARY SELF-CHECKOUT

ADMIN ACCESS
Kelvin Fong Wen Kiong
ROLE: ADMIN

Overview

Manage Users

Catalogue

Book List

Book Items

Borrow Books

Return Books

Manage Holds

AI Recommendations

Learning

Social

Sign Out

ADMIN CONTROL CENTER

Good day, Kelvin Fong Wen Kiong

Full catalogue and circulation access enabled for this session.

SIGNED IN:
adminaccount@kelvinfongwenkiong.com
ROLE: ADMIN

TOTAL ITEMS
All catalogue records
120

AVAILABLE NOW
Copies ready for lending
51

ACTIVE LOANS
Currently checked out
2

OVERDUE
Follow up required
2

Recent activity

BORROWER	BOOK	BORROWED	STATUS
Kelvin Fong Wen Kiong Admin - ID 102782287	Computer Networks Explained Barcode AUTO-74908896AD91	8 Apr 2026	Returned 10 Apr 2026
John Doe User - ID 535	Financial Markets & Institutions Barcode AUTO-717018190A9C	5 Apr 2026	Returned 8 Apr 2026
John Doe User - ID 535	Control Systems Introduction Barcode AUTO-1C089CA3258D	2 Apr 2026	Returned 5 Apr 2026
John Doe User - ID 535	Computer Networks Explained Barcode SWIK-20230006	1 Apr 2026	Returned 1 Apr 2026
Kelvin Fong Wen Kiong Admin - ID 102782287	Illustration Techniques Barcode AUTO-888188E78E44	27 Mar 2026	Returned 2 Apr 2026
Kelvin Fong Wen Kiong Admin - ID 102782287	User Experience Design 101 Barcode SWIK-20230004	27 Mar 2026	Returned 2 Apr 2026

Latest borrowing and return activity

Returned
Book Returned
"Computer Networks Explained" has been returned.

Figure 36: pop-up notification window

3.1.5 Borrow History

In order to ensure that users remain updated about the books they have borrowed from us, a history feature is provided by us. The books that users have borrowed, have borrowed and returned, or have not returned can be seen using this history feature.

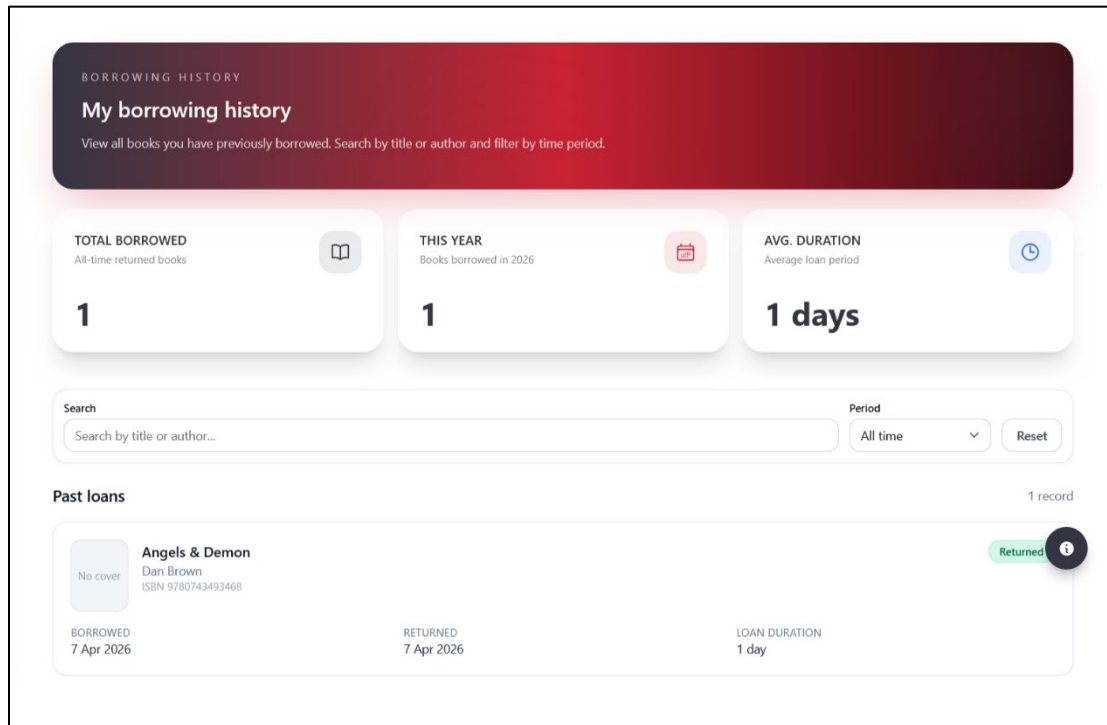


Figure 37: Borrow history page

3.1.6 AI Book Recommendation

For the suggested AI books, the recommended AI system is Google Gemini LLM. LLM refers to AI models that are created through deep learning techniques, relying on large data sets; therefore, it is highly appropriate for use as our AI tool.

By default, the LLM AI offers a wide range of functions, such as chat and content creation. However, we wish for this AI to have only specific functions, such as recommending books and suggesting reading order. To limit the functionality of our AI tool and prevent users from using it indiscriminately and wasting tokens, we have introduced a prompt restriction: the AI can only be used when users are searching for books; in all other circumstances, its use is prohibited.

We can see that our prompt will be having keywords for books, borrowing, and classifications, such as computer science and borrow.

Also it is now integrated to Learning path system, which will together respond to the user's prompt.

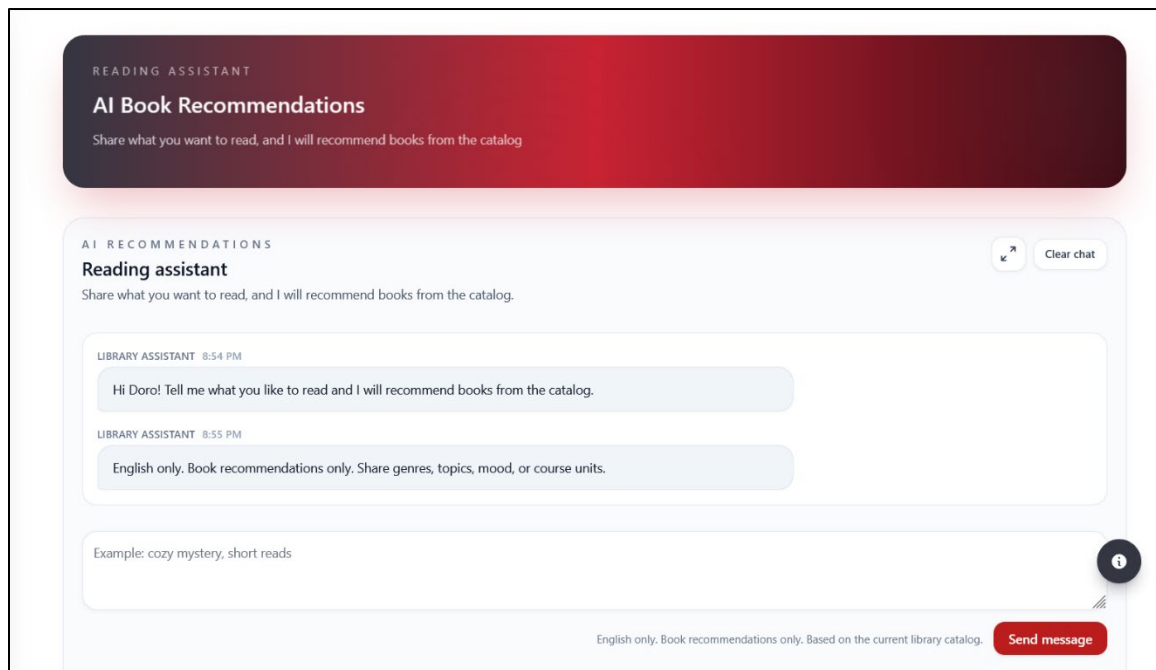


Figure 38: AI Book Recommendation

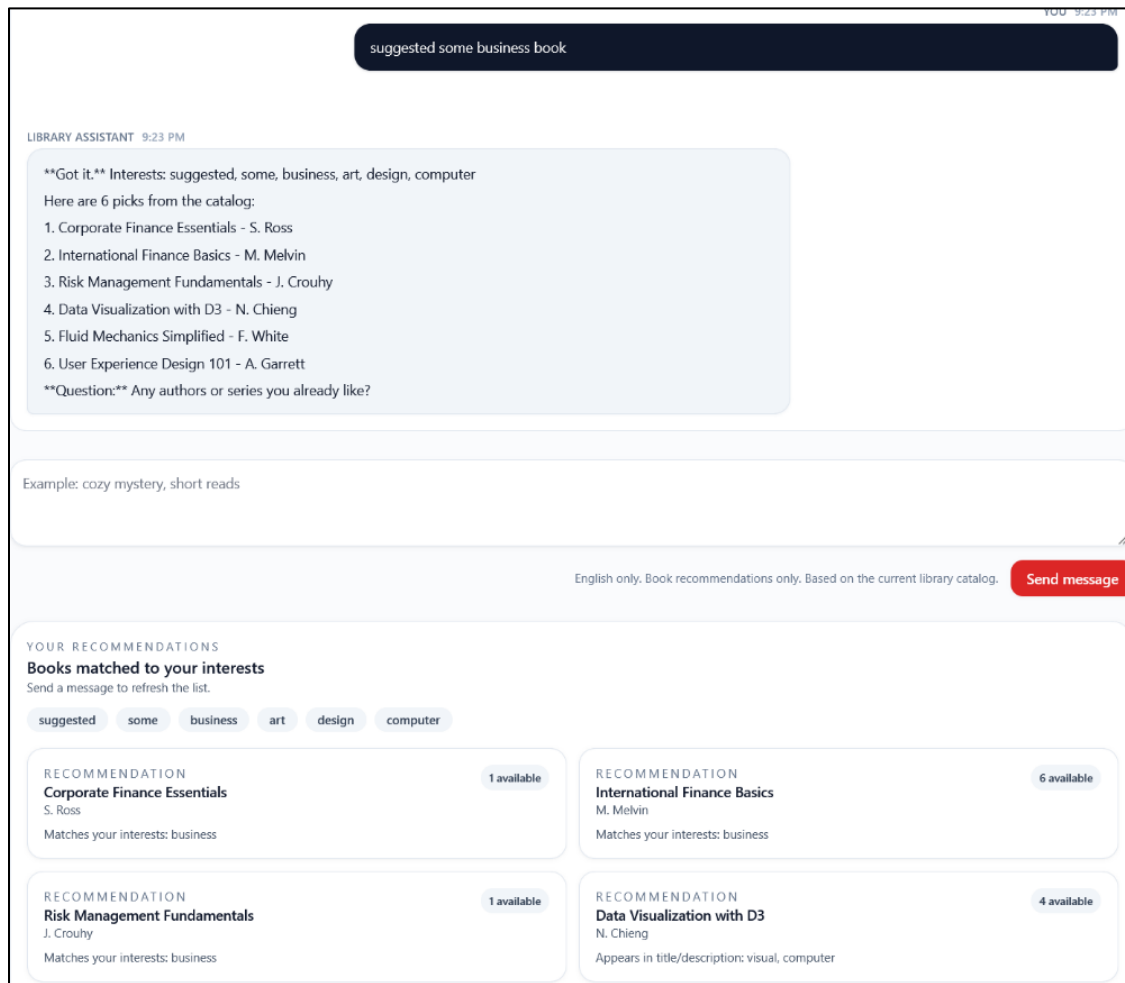


Figure 39: AI Book Recommendation Suggestion

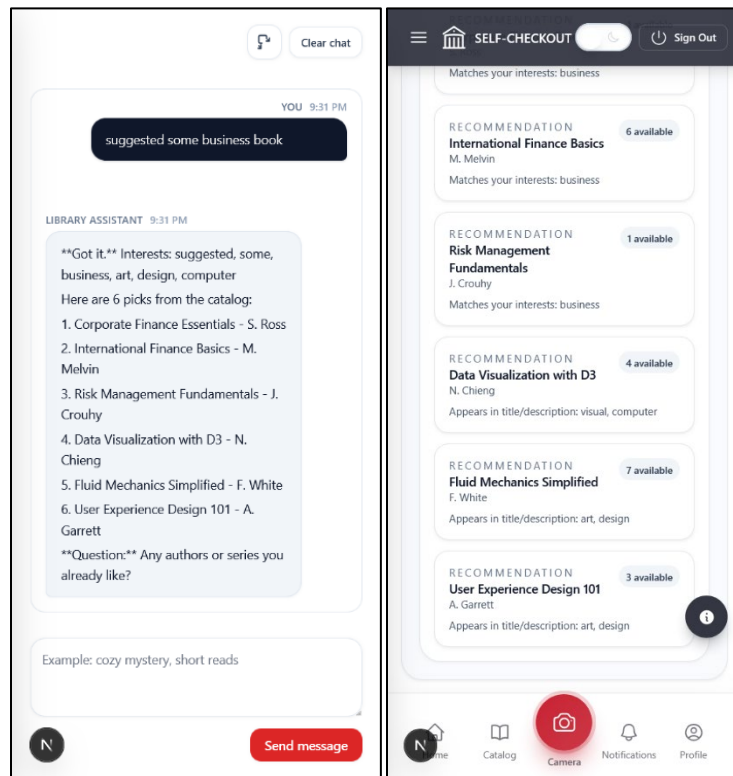


Figure 40: AI Book Recommendation (mobile)

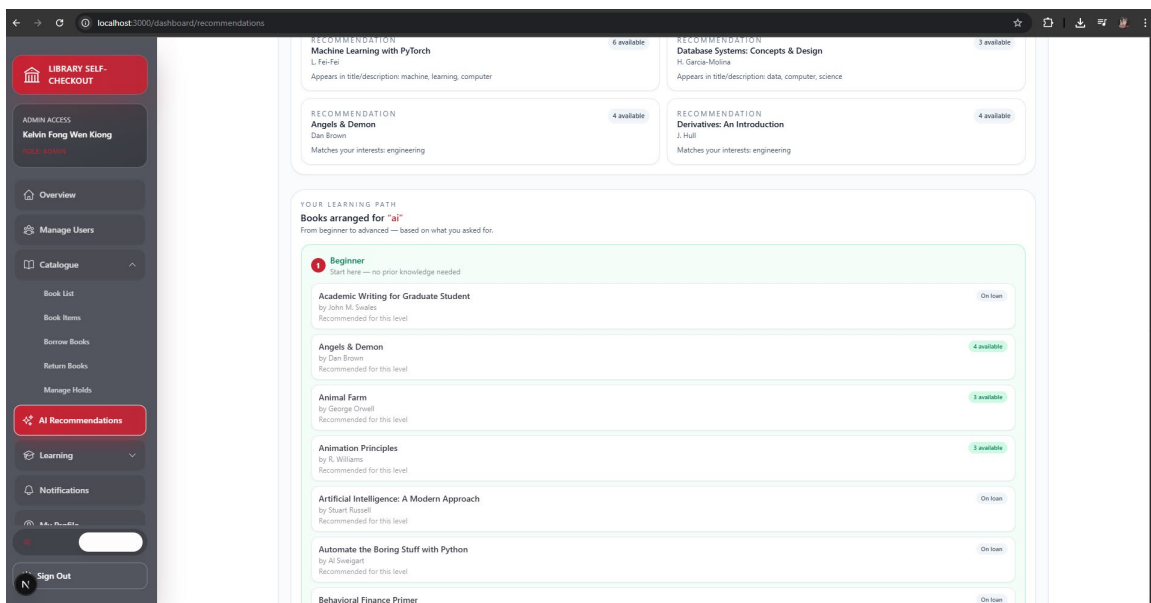


Figure 41: Learning path system integrate with AI book recommendation system

3.1.7 LinkedIn Learning

In addition to supporting users' personalized learning, we have also introduced the LinkedIn Learning tool. Initially, the idea was to suggest LinkedIn Learning modules and paths that would be appropriate for users by analyzing their needs through AI.

However, in relation to the LinkedIn account, since LinkedIn needs to be subscribed to, we cannot fully utilize the functions that LinkedIn offers. As a result of the limitations posed by LinkedIn itself, we are only able to do as much as create a page wherein its contents redirect to the LinkedIn account page.

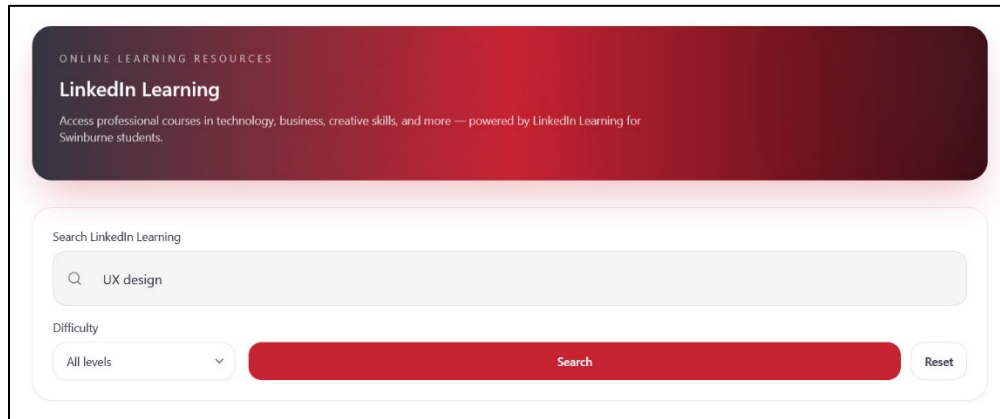


Figure 42: LinkedIn Learning Page

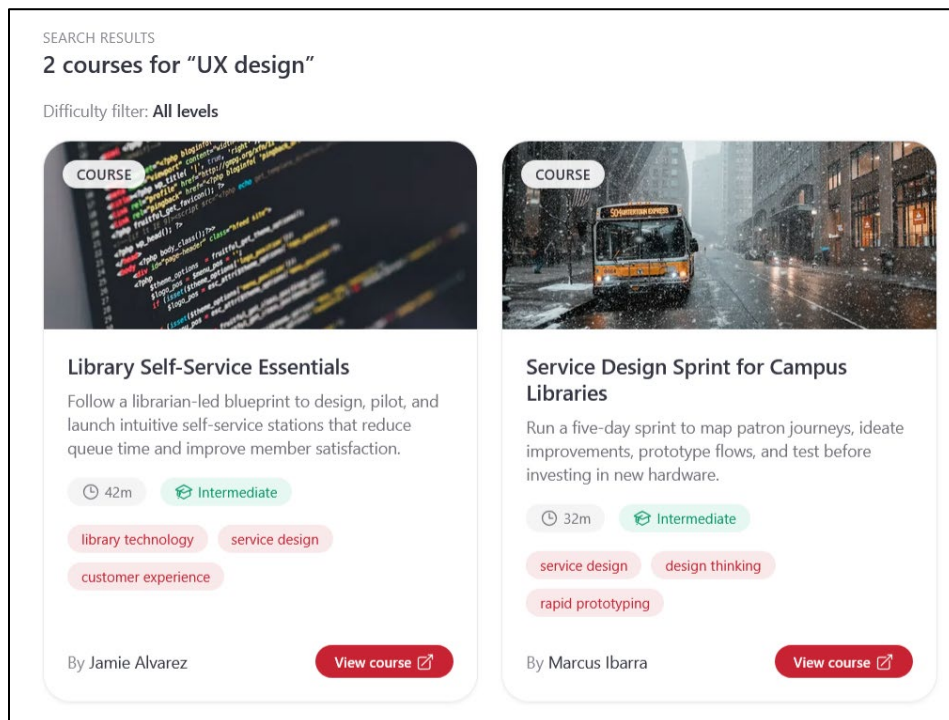


Figure 43: LinkedIn Topic

3.2 REPOSITORY AND WORKFLOW EVIDENCE

3.2.1 GitHub

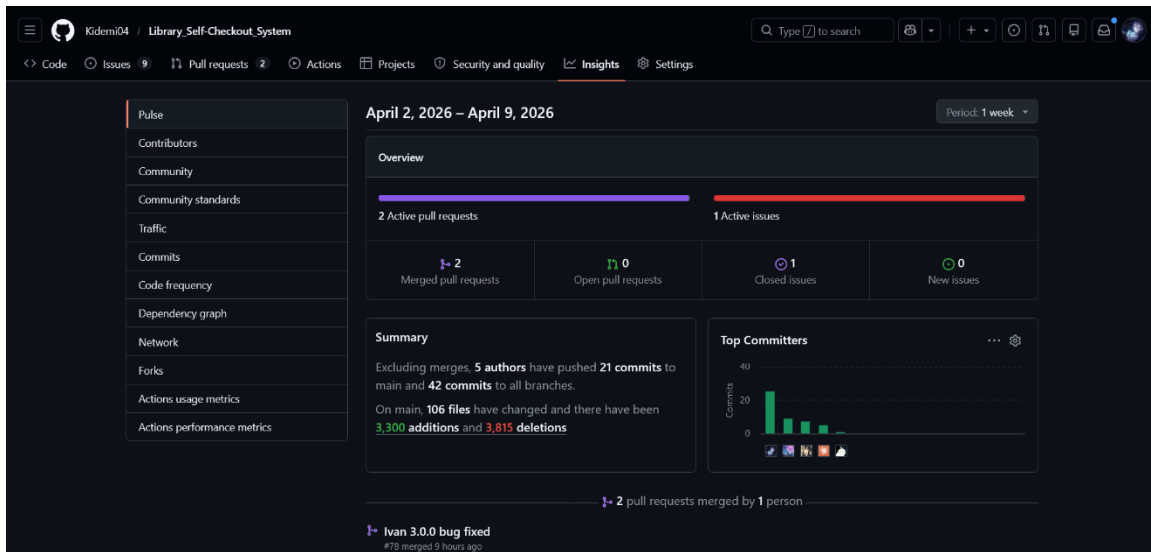


Figure 44: GitHub Insights

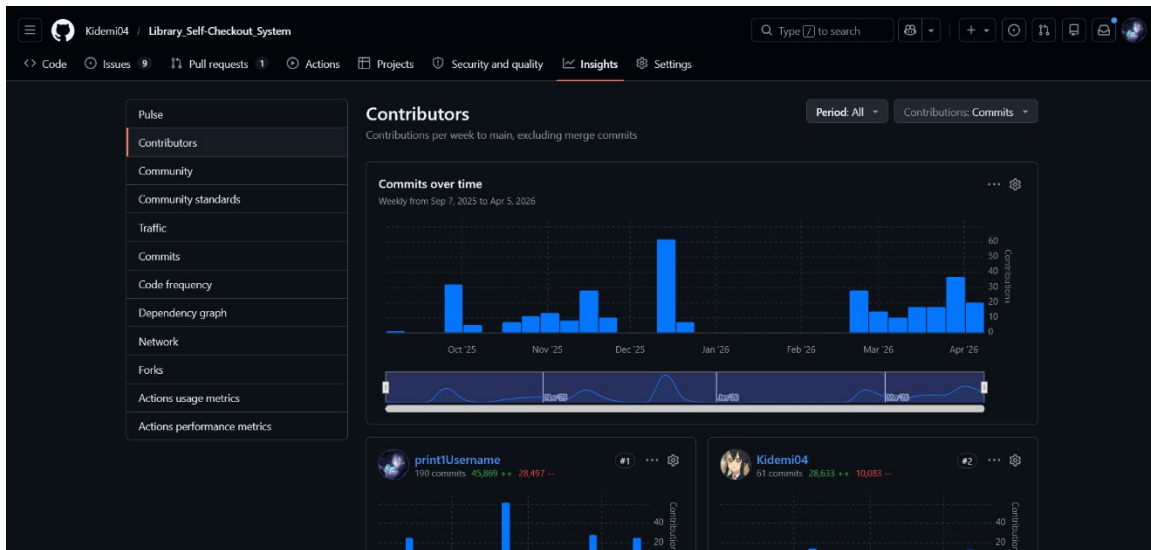


Figure 45: GitHub Contributors

Kidemi04 / Library_Self-Checkout_System

Code Issues 9 Pull requests 1 Actions Projects Security and quality Insights Settings

Actions New workflow

All workflows

Test Build Next.js in Ubuntu

Management

- Caches
- Deployments
- Attestations
- Runners
- Usage metrics
- Performance metrics

All workflows

Showing runs from all workflows

136 workflow runs

Workflow Event Status Branch Actor

q	Test Build Next.js in Ubuntu #165: Commit 6d6a5c7 pushed by print1Username	Ken-v2.7.1-Account-Interest	41 minutes ago	36s	...
pp	Test Build Next.js in Ubuntu #164: Commit f5e847f pushed by print1Username	Ken-v2.7.1-Account-Interest	46 minutes ago	37s	...
small bug fixed	Test Build Next.js in Ubuntu #163: Commit c525c90 pushed by kenn040502	Ken-v2.7.1-Account-Interest	Today at 12:15 AM	35s	...
learning path	Test Build Next.js in Ubuntu #162: Commit 3849a3f pushed by kenn040502	Ken-v2.7.1-Account-Interest	Today at 12:13 AM	34s	...
user role activity	Test Build Next.js in Ubuntu #161: Commit 4232052 pushed by kenn040502	Ken-v2.7.1-Account-Interest	Apr 9, 11:25 PM GMT+8	39s	...
own user activity	Test Build Next.js in Ubuntu #160: Commit 8d65d24 pushed by kenn040502	Ken-v2.7.1-Account-Interest	Apr 9, 11:05 PM GMT+8	38s	...

Figure 46: GitHub Actions

3.2.2 WhatsApp Chat

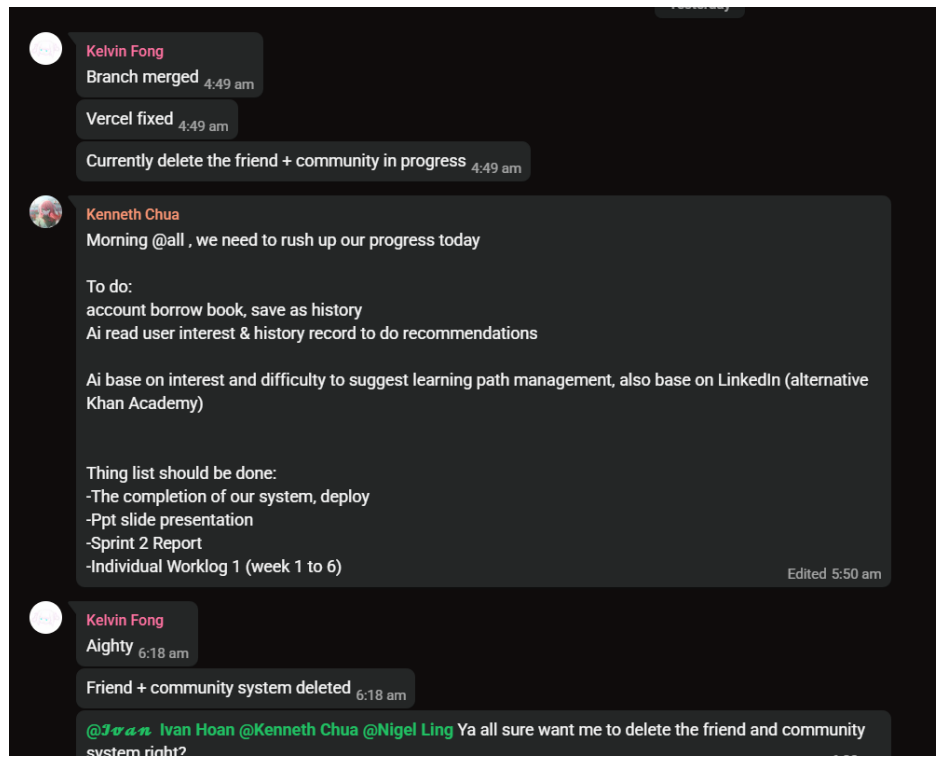


Figure 47: Each of the group members are discussing and reporting progress

3.2.3 Microsoft Teams

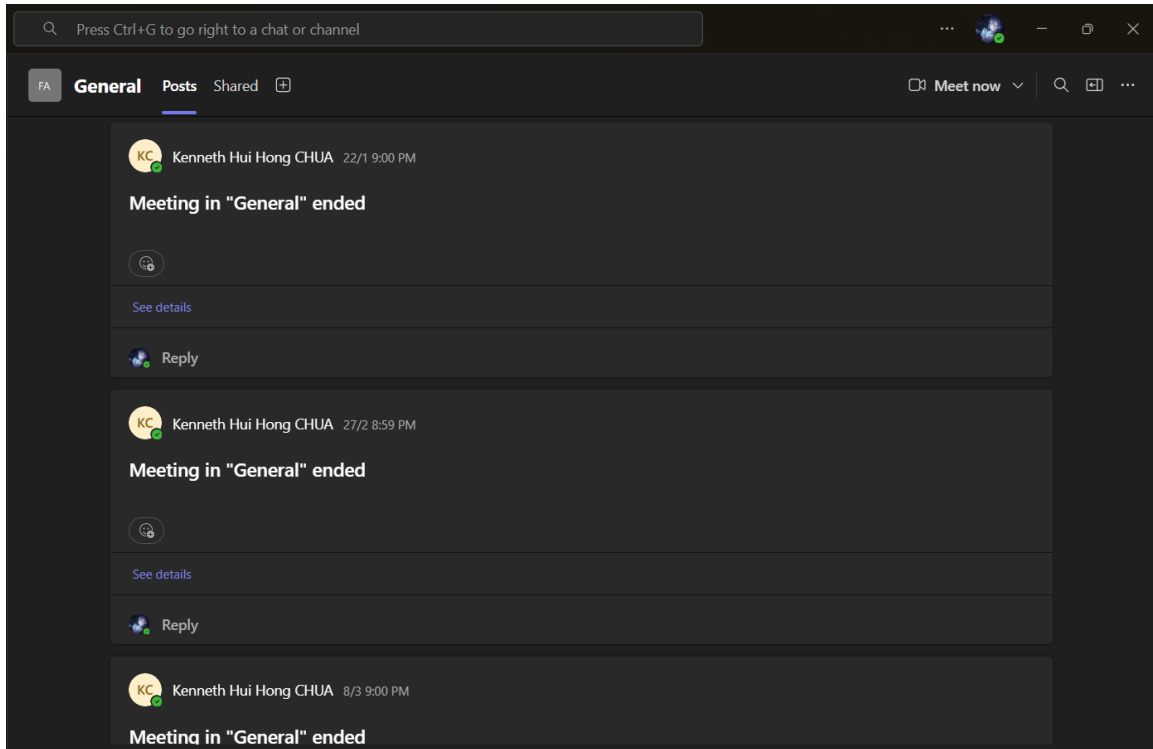


Figure 48: Evidences of group meeting (1/2)

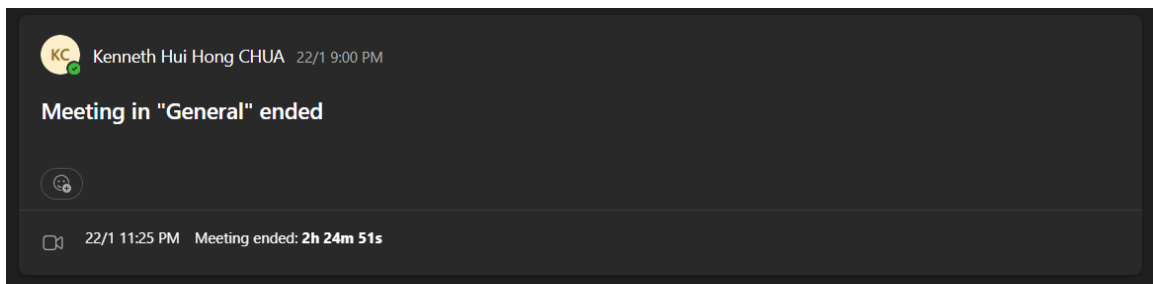


Figure 49: Evidences of group meeting (2/2)

3.2.4 Linear

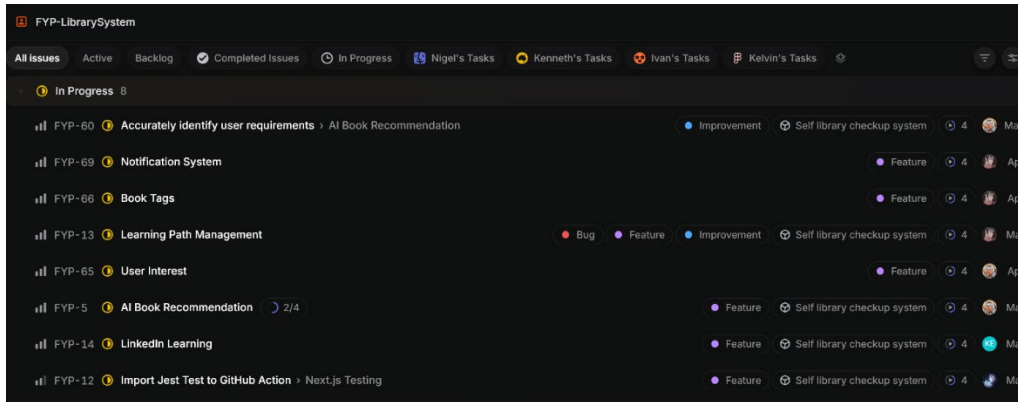


Figure 50: Linear - In Progress

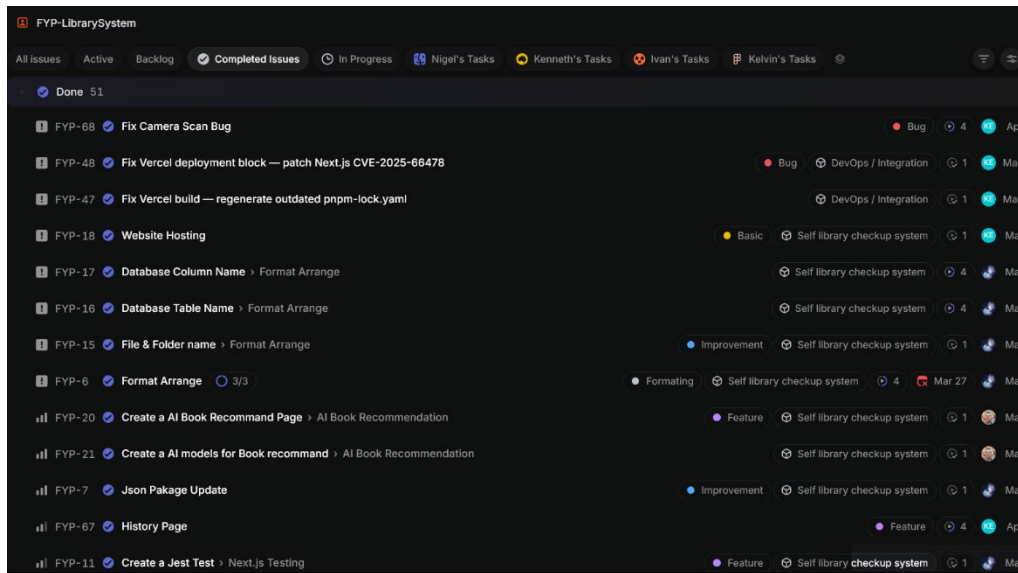


Figure 51: Linear – Done

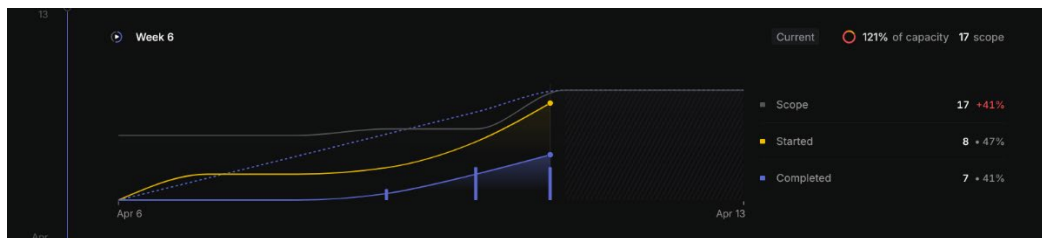


Figure 52: Linear - Cycle

4.2 FEEDBACK RECEIVED

During the sprint review, we received valuable feedback regarding the core features of the system, specifically focusing on the AI-powered book recommendations, learning path management, and LinkedIn Learning integration. The feedback centered on expanding the system's accessibility and enhancing the personalization of the recommendation engine:

4.2.1 Implementation of a Local LLM for Offline Capabilities:

A critical point raised was the need to support users who may not have a continuous internet connection. It was strongly recommended that we integrate a local Large Language Model (LLM). This would allow the system to continue processing data and providing offline users with AI-driven recommendations and learning path adjustments without relying on cloud-based processing.

4.2.2 Enhanced Personalization via Broad Online Data Tracking:

To make the user's personal data profile more robust and tailored to their specific interests, feedback suggested expanding how the AI tracks and gathers data. Instead of relying solely on in-system behavior, the AI should be capable of tracking broader online data footprints to build a highly personalized interest graph.

4.2.3 Leveraging Multimedia and Social Platforms for Knowledge Profiling:

Building on the data tracking feedback, it was specifically suggested to pull interest data from multimedia platforms such as YouTube, Instagram, and TikTok. The rationale behind this is that "books are knowledge," and in the modern digital landscape, knowledge consumption extends far beyond traditional reading. By analyzing a user's engagement with educational or topical content on video and social media platforms, the AI can generate much more accurate, cross-disciplinary book and learning path recommendations.

4.3 CRITICAL ANALYSIS OF PROGRESS

Sprint 2 aimed to fully complete the entire Library Self-Checkout system, including all core features like book borrowing, returning, notifications, reservations, and AI powered features like book recommendations and learning paths. The team made good progress overall, but a few major problems slowed things down and pushed the project behind schedule.

The biggest problem this sprint was the database migration. The team needed to rename all the database tables and columns to match the format the client requested. This was supposed to be done early in the sprint so the rest of the features could be built on top of the updated database. However, because the naming changes affected almost every part of the code, the team had to go through and fix queries one by one across the entire system. This took much longer than expected and caused delays for other features that were waiting on the database to be ready.

The Gemini AI rate limit was also a continuing problem. Because the team is using the free version of the Gemini API, it hits its usage limit very quickly during testing and development. When this happens, the AI chatbot and recommendation system stop responding completely, making it hard to properly test and show these features. This issue carried over from Sprint 1 and has not been fully solved yet.

The LinkedIn Learning integration was officially dropped this sprint. The feature requires an enterprise license that the team does not have, and the client has not responded to multiple follow up messages asking for clarification. Without a clear answer from the client, the team made the decision to stop pursuing this feature.

On a positive note, the team completed a number of important features this sprint including the notification system, reservation system improvements, AI book tagging, user interest profiling, and the full database migration. The team also removed the Friends and Community social features after agreeing that library users are unlikely to use them, which made the system simpler and easier to maintain.

Overall, the project is currently behind the original schedule going into Sprint 3. The main priorities for the next sprint are completing user acceptance testing, improving the accuracy of the AI recommendation system, and finishing any remaining features before the final deadline.

5 Retrospect (Critical review of the process)

5.1.1 What went well

The reservation system was successfully improved this sprint, with a new requirement added that staff must confirm a book has been returned before marking a reserved copy as ready for collection. This prevented premature reservation notifications from being sent to users. The notification system was also implemented smoothly, with all notification triggers functioning as expected across all user roles without requiring further debugging after the initial implementation.

The team maintained consistent communication throughout the sprint via WhatsApp and Microsoft Teams, with members generally responding within acceptable timeframes. Branch integration was handled efficiently, with all pull requests reviewed and merged without major disruptions to the shared codebase.

5.1.2 What did not go well

The database migration took significantly longer than anticipated. Because the client required all table names to follow PascalCase and all column names to follow snake_case, every query across the entire codebase had to be updated manually. This was a time-consuming process that delayed other features which depended on the updated database structure, and contributed to the project falling behind schedule.

The AI book tagging implementation did not produce the expected results. The initial plan was to give every book a unique descriptive tag, however the AI generated a different tag for each book which made categorization harder rather than easier. The team had to revise the approach and instead used the AI to map books into four fixed categories. This rework consumed additional time and highlighted the risk of relying on free-tier AI APIs for bulk operations, as rate limits were frequently hit during the process.

Task tracking on Linear was inconsistent throughout the sprint. Team members did not update their task statuses in real time, which meant the board did not accurately reflect actual progress during the sprint. This made it difficult to identify blockers early and led to a last-minute rush of status updates near the end of the sprint.

Some team members experienced delays completing their tasks due to other academic commitments during the sprint period. This affected the overall pace of development and meant that certain features were completed later than originally planned.

5.1.3 What can be improved

The team needs to establish a habit of updating Linear task statuses regularly, ideally at the end of each working session. This would give the team leader and other members a more accurate picture of progress and allow blockers to be identified and addressed earlier.

The Gemini API free tier limitations need to be addressed going into Sprint 3. The current approach of using a single free-tier API key is not sustainable for a system that depends on

AI for core features. The team should explore using multiple API keys in rotation or switching to a more capable model to avoid rate limit issues during testing and development.

Communication response times could also be more consistent. While most messages were responded to promptly, there were occasional delays in responses on WhatsApp which sometimes slowed down decision making. Agreeing on a maximum response window for urgent queries would help maintain momentum.

5.1.4 Action item to be done

- All team members to update Linear task statuses at least once per day during active development periods
- Investigate alternative AI API solutions or a paid tier to resolve the Gemini rate limit issue before Sprint
- Complete remaining Jest tests to cover core API endpoints and database query functions
- Conduct user acceptance testing with real users in Sprint
- Finalise any outstanding features before the Sprint 3 deadline

5.1.5 General observations from Sprint 2

Overall, Sprint 2 demonstrated that the team is capable of delivering complex features under time pressure, but process discipline needs improvement. The biggest recurring issue across both sprints has been the gap between actual progress and what is reflected in the task tracking system. When Linear is not updated in real time, the team loses visibility over the sprint and risks missing blockers until it is too late to address them comfortably.

The decision to remove the Friends and Community social system was a good example of the team making a pragmatic scope decision based on user needs rather than sticking rigidly to the original plan. This kind of critical thinking about scope will be important in Sprint 3 as the team prepares for user acceptance testing.

The team's code of conduct, which includes completing tasks on time, communicating proactively, and respecting each other's contributions, was generally followed throughout the sprint. The main area where conduct fell short was task tracking discipline, which will be a focus going forward.

5.1.6 Risk Registry and Ethical Considerations

No.	Risk	Likelihood	Impact	Rating	Responsible	Mitigation & Current Status
1	Progress delay due to uneven skills and procrastination	High	High	High	Kelvin, Kenneth, Nigel	Mitigated through direct monitoring and clearer task assignments. Some delays occurred this sprint due to academic commitments.
2	LinkedIn Learning API credentials not received from client	High	High	High	Kenneth, Kelvin	Feature put on hold this sprint as the integration requires an enterprise license that the team does not currently have access to. The team will revisit this if client clarification is received in a future sprint.
3	AI API instability or rate limit exceeded	High	High	High	Kenneth	Free tier Gemini API hit rate limits frequently this sprint, causing AI features to become unresponsive.

						To be addressed in Sprint 3 by exploring alternative solutions.
4	Insufficient test coverage leading to undetected bugs	Medium	High	Medium	Ivan, Kelvin	28 tests passing with zero failures. Core API endpoints and database queries still have no test coverage. To be expanded in Sprint 3.
5	Supabase schema changes breaking existing functionality	Low	High	Medium	Ivan, Nigel	Database migration to PascalCase and snake_case completed this sprint. All affected queries updated and verified. Risk now reduced.
6	Deployment pipeline instability blocking production releases	High	High	Low	Kelvin	Two deployment issues resolved this sprint. Pipeline currently stable.
7	Task tracking discipline	High	Medium	Medium	All members	Linear not updated in

	leading to poor sprint visibility					real time throughout Sprint 2. Action item raised to address this in Sprint 3.
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Ethical Considerations:

The system collects and stores personal data including student names, email addresses, student IDs, borrowing history, and user interest preferences. The team is committed to handling this data in accordance with the Australian Privacy Principles (APPs) under the Privacy Act 1988. Specifically, the system only collects data that is necessary for the library system to function, data is not shared with third parties without user consent, and access is controlled through role-based authentication to ensure users can only access their own data.

The AI recommendation system raises additional ethical considerations around transparency and accuracy. Users should be aware that recommendations are generated by an AI model and may not always reflect their actual preferences accurately. The team has implemented guardrails to prevent the AI from responding to off-topic or inappropriate queries, and recommendations are grounded in the actual library catalogue rather than external data sources.

The removal of the Friends and Community social features also reduces the amount of social interaction data the system collects, which is a positive step from a privacy perspective.

6 Lessons learned (Critical review of SPRINT one experience and future plan)

6.1.1 Lessons learned

We learned that maintaining a consistent format and structure throughout the project is very important. Inconsistent naming conventions, coding styles, and documentation led to confusion and increased the likelihood of errors. Establishing clear standards early would have improved efficiency and reduced mistakes. The front-end of the system still has usability issues. Some features are not intuitive, and the overall interface can be confusing for users. This highlighted the importance of prioritizing user experience (UX) design and conducting usability testing earlier in the development process. Integrating AI components introduced several challenges. For example, in the prompt injection detection task, it was difficult to ensure the AI consistently identifies malicious or irrelevant inputs. Since the LLM is a cloud-based service, we have limited visibility into how it processes information or whether it accesses external knowledge sources. This lack of transparency makes it harder to control and evaluate its behaviour. We also faced challenges in using AI to categorize books into difficulty levels (e.g., beginner, intermediate, advanced) within the learning path system. It is unclear how the AI determines the difficulty level, especially without explicitly analysing the full content of the book. This raised concerns about the reliability and accuracy of AI-generated classifications. Overall, this project helped us realize that AI is not always reliable or “intelligent” in the way we expect. It requires proper prompting, validation, and sometimes additional logic or rules to ensure acceptable performance.

6.1.2 To-do and future improvements

In future iterations, we plan to optimise the way the AI operates so that the recommendation system can better understand and respond to users’ needs. This includes exploring more effective approaches for implementing AI recommendations in the long term, ensuring both reliability and scalability. Additionally, we aim to evaluate alternative methods to replace or enhance the current LinkedIn Learning integration. From a technical perspective, we intend to experiment with different AI models, including both local LLMs and cloud-based LLMs, to compare their performance, cost, and flexibility. The user interface will also be improved, particularly by optimising the navigation bar to enhance usability and user experience. Furthermore, some system functions are still not fully integrated, and completing these integrations will be a priority. In terms of testing and validation, we plan to conduct User Acceptance Testing (UAT) using Google Forms to collect structured feedback. We will also carry out physical testing sessions with students and librarians to better understand real user needs and identify areas for improvement.